SAINT MICHAEL COLLEGE OF CARAGA

THE INTERPLAY OF 21ST-CENTURY COMPETENCIES AND LINGUISTIC COMPETENCE: A CORRELATIONAL ANALYSIS

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The Faculty of
College of Teacher Education

In Partial Requirements for the Degree
BACHELOR OF SECONDARY EDUCATION MAJOR IN ENGLISH

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SAINT MICHAEL COLLEGE OF CARAGA
College Of Teacher Education

CERTIFICATE OF RESEARCH APPROVAL

This thesis entitled **"**The Interplay of 21st-Century Competencies and Linguistic Competence: A Correlational Analysis**"** presented and submitted by BETHANY MAE P. MONTILLA in partial fulfillment of the requirements for the degree of Bachelor of Secondary Education Major in English is hereby accepted and recommended for Oral Examination.

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ABSTRACT

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I. OBJECTIVES

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the Study

Education today requires more than knowing facts and completing tasks. Students must be able to communicate clearly, think carefully, work well with others, and create new ideas. These abilities, known as the 4Cs of 21st-century skills—communication, collaboration, critical thinking, and creativity—are now important for success in school, at work, and in daily life (Thornhill-Miller et al., 2023; Martínez & Bonilla, 2024)

At the center of these skills is linguistic competence, or the ability to use language correctly and effectively. Linguistic competence includes phonology (the sounds of language), vocabulary, word formation, sentence structure, meaning, and correct spelling. Students who have weak control of these areas often find it difficult to understand lessons, express ideas, solve problems, or participate in group activities (Qizi, 2024; Mytsenko, 2024). Strong language skills, on the other hand, help students speak clearly, organize their thoughts, and interact confidently with others.

Many international studies show that language skills and 21st-century skills are closely connected. Ünaldı and Yüce (2021) explained that understanding grammar and forming correct sentences helps students think more critically. Susoy (2024) found that students who analyze information well also perform better in reading. Other researchers observed that practicing language improves accuracy and adaptability (Alkhrisheh et al., 2021), while the creative use of words supports innovation and problem-solving (Sastromiharjo & Hadianto, 2024). Shakirova et al. (2019) added that combining language practice with critical thinking increases creativity. Studies by Shadiev and Wang (2022) and Mytsenko (2024) also showed that strong language

proficiency allows learners to communicate and collaborate across cultures. Together, these findings suggest that developing linguistic competence naturally strengthens the 4Cs.

Philippine studies support the same idea. Cabico and Sales (2023) reported that grammar proficiency improves fluency and teamwork. Galang et al. (2024) found that English proficiency predicts better performance in Purposive Communication. Caldoza et al. (2024) showed that reading comprehension helps students think more critically and reflect on what they learn. Azarcon-Sanchez et al. (2024) noted that English teachers actively promote creativity and critical thinking in their classes. Herradura Basilan and Paula Flor (2024) found that communicative activities help build collaboration among students, while Kadatuan et al. (2024) confirmed that communicative competence improves when learners participate actively in language tasks.

Even with these findings, many colleges still have difficulty connecting language learning with the development of the 4Cs. Several studies noted that language subjects are often taught separately from activities that build communication, collaboration, creativity, and critical thinking (Pardede, 2020; Caldoza et al., 2024). Because of this, students may complete written requirements but still struggle to explain their thoughts, work with peers, or produce original outputs. This situation shows a gap between what students learn in language classes and how they apply these skills in real-life learning tasks (Azarcon-Sanchez et al., 2024;Mae et al., 2024)

A similar concern can be seen at Saint Michael College of Caraga (SMCC), especially among first-year students from the College of Teacher Education (CTE) and the College of Arts and Sciences (CAS). Some students have basic English knowledge but find it hard to express ideas clearly, organize their thoughts, or participate actively in group work. Others can work alone but struggle when asked to think deeply, communicate complex ideas, or create original outputs. These challenges suggest that their level of linguistic competence may be affecting how well they develop the 4Cs.

Although many international and local studies have explored language skills and 21st-century skills**,** very few have examined how specific components of linguistic competence—such as phonology, vocabulary, grammar, and meaning—relate to the 4Cs among first-year Filipino college students. No study has yet focused on this relationship within the context of SMCC, where concerns about communication, collaboration, creativity, and critical thinking continue to appear in classroom situations.

Because of this gap, it is important to investigate how linguistic competence influences the 4Cs among SMCC first-year college students.

This study examines the relationship between students' linguistic competence and their communication, collaboration, creativity, and critical thinking skills. The findings will benefit students of Saint Michael College of Caraga by strengthening their language proficiency and essential 21st-century competencies. Teachers handling English and language-related subjects may use the results to enhance instructional strategies and learning activities that integrate linguistic competence with the 4Cs. School administrators may utilize the findings to support curriculum planning, instructional supervision, and professional development initiatives. Moreover, the College of Teacher Education and the College of Arts and Sciences may use the results for curriculum review and program enhancement to ensure that language instruction and academic programs are aligned with the demands of 21st-century education. Overall, the study contributes to the development of linguistically competent and globally competitive Filipino graduates.

1.2 Review Related Literature

This chapter examines new studies on language skills— phonology, morphology, syntax, semantics, and orthography—as well as the 4Cs of 21st-century skills: critical thinking, communication, collaboration, and creativity. It focuses on research that shows how language knowledge connects with the skills needed in school and work today. Many studies also show how technology helps both language learning and the 4Cs.

Linguistic Competence

Canale & Swain (1980) conceptualized linguistic competence as an individual's knowledge of the formal aspects of language, including grammar, vocabulary, and phonology, which enable accurate and meaningful language use. Within their theory of communicative competence, linguistic competence serves as the foundational component that underpins effective communication across contexts. Supporting this theoretical perspective, Ergasheva (2025) explained that linguistic competence represents a speaker's underlying knowledge of language systems such as syntax, phonology, morphology, and semantics, which collectively allow individuals to comprehend and produce language appropriately. This conceptualization establishes linguistic competence as a core construct in both theoretical and applied linguistics.

Empirical studies further clarify the components and functions of linguistic competence. Wahyuni et al. (2022) emphasized that linguistic competence enables learners to understand and express meaning clearly through appropriate use of grammatical structures, vocabulary, and pronunciation, making it essential for effective communication in educational and social environments. Similarly, Świątek et al. (2023) identified grammar, pronunciation, and vocabulary as the primary components that allow learners to construct accurate sentences and communicate

effectively. These findings highlight that linguistic competence is a multi-component construct rather than a single isolated skill.

Research in second language learning contexts demonstrates the role of linguistic competence in developing receptive and productive language skills. Singhkum and Patanasorn (2022) found that learners' listening comprehension is significantly influenced by their control of pronunciation, sentence structure, and meaning, indicating the interrelated nature of linguistic components. In writing contexts, Y. Wang and Xie (2022) reported that both linguistic and discourse competence contribute to improved writing accuracy and coherence among students. Likewise, Younis and Shah (2023) observed that communicative-oriented textbooks enhance learners' linguistic competence by integrating grammatical knowledge with meaningful language use, while Mao (2023) stressed the importance of teaching learners how to apply language skills in real-life situations.

Studies also suggest that linguistic competence is shaped by learners' social backgrounds and learning environments. Davydova (2022) revealed that social identity and linguistic background influence how learners acquire and use English, affecting their overall language development. In professional contexts, Sebahat Gözüm et al. (2020) emphasized that effective linguistic competence is essential across occupations, including healthcare, where clarity and accuracy of language use directly affect service quality. These findings indicate that linguistic competence extends beyond structural knowledge and is influenced by cultural and contextual factors.

Despite extensive international research, gaps remain in understanding how linguistic competence relates to broader educational competencies. Chan et al. (2023) found that bilingual learners' language systems develop interdependently, suggesting the need for longitudinal and context-sensitive studies. Meanwhile, Yu and Dong (2022) demonstrated that technology-

enhanced learning environments can support the development of linguistic competence through interactive and learner-centered instruction. However, in the Philippine context, few studies have explicitly examined the relationship between linguistic competence and the 21st-century competencies of communication, collaboration, critical thinking, and creativity.

In summary, linguistic competence is a fundamental element of language learning and effective communication. Grounded in communicative competence theory and supported by empirical research, it connects linguistic knowledge with real-life language use. Given the limited local studies linking linguistic competence with the 4Cs, further research is necessary to better understand its role in supporting holistic learner development in contemporary educational contexts.

Phonology

Phonology is a core component of linguistic competence, encompassing both segmental features and suprasegmental features such as stress, rhythm, and intonation. Explicit phonological instruction significantly enhances learners' ability to perceive and produce sounds accurately. For example, Shukurullayeva and Qizi (2024) found that phonetics-based instruction meaningfully improves pronunciation and sound discrimination, reinforcing the need for phonology-informed pedagogy. This is supported by Suzukida and Saito (2022) who demonstrated that segmental and suprasegmental features jointly shape L2 English pronunciation, indicating that phonological competence contributes not only to sound production but also to overall communicative effectiveness.

Building on this foundation, research on phonological awareness (PA) highlights its role in early literacy development. PA enables learners to identify and manipulate sound units such as syllables and phonemes, skills essential for decoding and reading. In the Philippine context, Abdon and Barrios (2022) showed that a structured mother-tongue PA intervention significantly improved

kindergarten learners' phonological awareness, letter knowledge, and even early English reading abilities. These findings support Cummins's Linguistic Interdependence Hypothesis, suggesting that strengthening phonological skills in L1 can positively transfer to L2 literacy. This underscores the importance of mother tongue–based multilingual education, especially in linguistically diverse communities.

As literacy-related aspects of phonology develop, research on pronunciation emphasizes the importance of integrating both segmental and suprasegmental instruction. While learners often improve segmental accuracy more quickly, suprasegmental features play a crucial role in intelligibility and fluency. Luchini and Kickhöfel Alves (2025) found that suprasegmental instruction enhances prosodic awareness and contributes to more holistic pronunciation gains. Similarly, in a study of adult EFL learners, Rodríguez et al. (2025) reported that improvements in rhythm, stress, and intonation increased learners' fluency, expressiveness, and overall communicative confidence. This aligns with Aguion (2022) call for an intelligibility-oriented approach to pronunciation teaching in the Philippines, emphasizing communicative clarity over native-like accuracy.

As pronunciation pedagogy evolves, technological advancements have expanded its possibilities. Automatic Speech Recognition (ASR) systems now offer real-time, objective feedback that supports sustained pronunciation practice. Sun (2023) found that ASR-based training improved L2 learners' speaking performance by encouraging self-correction and iterative learning. Complementing this, a meta-analysis by Ngo et al. (2024) reported that ASR tools produce moderate and consistent improvements in pronunciation, particularly when paired with explicit corrective feedback. Moving toward more advanced models, Sungkur and Shibdeen (2025) developed a generative AI–enhanced pronunciation trainer that provides instantaneous feedback through ASR and TTS integration, showing strong potential for personalized and multilingual instruction. Duris et al. (2025) similarly argue that AI technologies enable scalable and

autonomous pronunciation practice, an idea supported by Liang and Fryer (2024) empirical findings that learners effectively engage with ASR tools in independent learning contexts.

Alongside these technological advancements, scholars have increasingly recognized that phonology interacts with other linguistic subsystems in shaping literacy and language development. Kim (2023) demonstrated that phonological awareness, morphological awareness, and orthographic pattern recognition collectively account for substantial variance in reading performance. This is echoed in findings by Zhang et al. (2021) who reported that morphological awareness supports learners' ability to activate phonological and orthographic representations in L2 reading. Furthermore, Li and Chen (2024) observed that learners' reliance on orthographic or phonological cues varies by proficiency, with more advanced readers integrating both systems fluidly. Intervention studies by Shen et al. (2024) also show that training in phonological and morphological skills strengthens reading outcomes, while cross-script research such as Young-Scholten et al. (2024) illustrates that L1 orthographic systems can shape L2 phonological development. Together, these findings reinforce the importance of integrated approaches to literacy instruction.

These interactions are especially visible in multilingual environments such as the Philippines, where L1–L2 transfer shapes learners' phonological development Sales (2022) for example, found that Cebuano-speaking junior high students experience difficulty with English vowels that do not align with Cebuano phonology. Sales (2022) documented similar transfer effects among Hiligaynon speakers, including both segmental and suprasegmental influences on English pronunciation. Beyond error-based perspectives, Dumanig et al. (2025) observed that some non-target-like phonological patterns among Cebuano-Visayan learners can aid meaning-making in academic discourse, illustrating the complex role of phonology in multilingual communication. Prosodic variation across Philippine English varieties also reflects strong L1

influence. Dumanig et al. (2025) found that acrolectal, mesolectal, and basilectal sociolects of Philippine English differ along a stress-timed to syllable-timed continuum. Teacher-related factors likewise shape phonology instruction, as Dumali and Bayan (n.d.) reported that limited teacher training and low confidence in teaching suprasegmentals hinder effective classroom implementation.

Although much of the research on phonology centers on literacy and pronunciation, emerging scholarship suggests broader relevance to 21st-century skills, particularly communication, collaboration, critical thinking, and creativity. Prosodic mastery enhances intelligibility and supports clearer interpersonal communication, an essential element of the 4Cs (Rodríguez et al., 2025). ASR-based tools also provide opportunities for peer-supported learning, which strengthens collaboration (Ngo et al., 2024). Additionally, phonological awareness interventions—especially in L1—promote metalinguistic reflection, encouraging learners to analyze how sound systems operate across languages, thereby supporting critical thinking (Abdon & Barrios, 2022). Creative exploration is likewise fostered by AI-based pronunciation tools, such as those designed by Sungkur and Shibdeen (2025) which allow learners to experiment with pitch, rhythm, and expressive variation.

Despite these promising directions, two significant gaps remain in existing research. First, little empirical work has investigated how phonological competence—especially suprasegmental proficiency—contributes to the development of 21st-century competencies among Filipino tertiary learners. Although some studies link prosody to communicative competence, its role in collaboration, critical thinking, and creativity has not been systematically examined. Second, althugh ASR and AI technologies show strong potential for pronunciation learning, few tools have been designed to fit the linguistic realities of Filipino learners, particularly those involving L1

phonological systems, Philippine English prosody, and local sociolinguistic variation. Research on integrating such tools into higher education curricula also remains scarce.

Given these gaps, the present study—examining the relationship between phonological competence and the 4Cs among Filipino tertiary students—offers a meaningful and timely contribution. By connecting phonology with higher-order competencies and localizing technological approaches to pronunciation learning, the study can advance both theoretical understanding and pedagogical practice in Philippine higher education.

Morphology

Morphology is the study of word structure and how the most minor units of meaning, called morphemes, combine to form words. It plays a vital role in literacy and language learning by helping learners understand how words carry meaning. Morphological awareness is described as the ability to recognize, break down, and manipulate morphemes to decode and construct new words (Altheneyan, 2025). While early research focused on children's literacy, more recent studies have extended the concept to adult and second-language learners, emphasizing its importance in higher education and multilingual settings.

Researchers have found that morphological awareness comprises two key steps: recognizing and manipulating shapes. Finding prefixes, suffixes, and base words is part of recognition. Manipulation means using this information to make or understand new words. Liu et al. (2022) found that studying morphemes makes it easier for students to read difficult material. Similarly, Kristensen et al. (2023) showed that students can infer the meanings of new words when they know how words are put together. These studies show that morphology helps people learn new words and understand their meanings by associating word shapes with their meanings.

Building on evidence that morphological awareness supports vocabulary development and reading comprehension, morphological knowledge is likewise beneficial in instructional contexts. Naser and Gandhi (2022) showed that teaching students to examine word parts makes it easier for them to put words together and understand their meanings, helping them read and write more independently. Activities in the classroom, such as finding affixes, learning about word families, and examining derivations, can help students learn new words and better understand what they read. These results make it clear that explicit morphological instruction should be part of second-language programs for both beginners and college students.

Recent studies have also examined how teaching morphology improves comprehension and vocabulary. Learning morphology helps students not only recognize words but also understand them better by connecting words with their meanings. Giazitzidou et al. (2025) report that practicing morpheme use improves students' reading and spelling. They also found that morphological awareness supports vocabulary knowledge and reading comprehension. These studies show that teaching morphology through word-building, affix recognition, and compound-word analysis can help learners improve their vocabulary and understanding.

Cross-linguistic studies show that grammar differs in different languages, which affects how people learn and use it. According to Tafese (2022) Oromo-speaking children have special problems learning inflectional grammar, which slows down their reading and writing growth. Mizan et al. (2023) also showed that knowing Arabic morphemes improves both communication and understanding. Shigini and Mapunda (2023) show that the way the Kemunasukuma people's personal names are put together shows how they see their culture and traditions. These instances show that morphology is both universal and culturally specific, with different effects across social and linguistic settings.

Additionally, technology has made it easier to learn about and use morphology in more situations. According to Muthee et al. (2022) morphological analysis supports natural language processing (NLP) in artificial intelligence by identifying root words that help computers better understand text. Bollmann and Søgaard (2021) said that machine learning systems struggle with languages with high grammatical richness, suggesting that we need better computer models. In education, digital tools such as live word analysis and mobile apps make learning morphology more fun and easier for students.

To sum up, the growing body of studies on morphology shows that it is used for a lot more than just teaching early reading. Morphological awareness is an important part of language, education, and technology that helps people learn new words, read and write in multiple languages, and use language correctly. Including it in multilingual and tertiary education connects theory and practice, giving students the metalinguistic awareness they need for school, work, and digital communication. Future studies should examine how morphological training and technology-based tools can work together to help students of all kinds understand what they read and learn new words.

Lexical Competence

Lexical competence is a part of communicative competence that refers to a person's ability to understand, recall, and use words appropriately in different situations. It involves more than just knowing the meaning of a word—it also includes understanding its form, common pairings (collocations), and correct use in context. According to López-Pérez et al. (2023), lexical competence may be viewed as a premise of effective communication because the extent of vocabulary knowledge, the number of words known by the learner, and comprehension, the level of language proficiency among the learner, are what determine the degree of proficiency within the whole language. Zare et al. (2021) further explain that lexical richness directly affects fluency

and communicative accuracy; thus, vocabulary development is a core aspect of academic and professional success.

Lexical competence has three dimensions: breadth, depth, and use. The breadth refers to the number of words that a learner knows, and depth is the level of information that the learner has of the form, meaning, and usage of a word (Shandra, 2021). In turn, use implies the possibility of applying lexical knowledge in real communicative situations through collocations and idiomatic expressions, as well as the context-specific use of words (Kaur, 2020; Khan & Anjum, 2023). All these dimensions reflect the cognitive and practical aspects of vocabulary acquisition and align with the 4Cs of 21st-century skills: communication, critical thinking, creativity, and collaboration. One of them is that the broad and rich vocabulary allows learners to share their multidimensional ideas, remain textually, and collaborate in different settings.

Studies show that various teaching strategies can strengthen vocabulary learning. Nishonova (2024) found that using the English language teaching process can not only help students acquire new vocabulary but also build creativity and awareness of context. Similarly, Kudryashova et al. (2020) found that contextual and meaningful exposure to vocabulary promotes long-term memory compared to frequency-based practices. Shandra (2021) notes that interactive vocabulary activities, metacognition reflection, and semantic mapping are active learning strategies that help learners move up the interlanguage continuum, converting passive knowledge of words into active competence.

Moreover, metacognitive awareness plays a key role in vocabulary learning. According to Kaur (2020), students who reflect on the vocabulary-learning method learn better and retain it longer when they do so consciously. Khan and Anjum (2023) reinforced the importance of lexical competence by showing that cognitive associations and word networks help learners form meaningful connections and use both their first and second languages effectively. These findings

highlight the importance of incorporating reflection, creativity, and context in teaching to make instruction learner centered.Lexical competence also extends beyond the classroom. In professional communication, vocabulary control is vital. Liu and Dou (2023) found that interpreters with strong lexical skills perform better under cognitive pressure because they can select precise words quickly. Shandra et al. (2022) also showed that context-based vocabulary teaching in the information technology field provides learners with lexical materials essential for their professional development. These studies emphasize that vocabulary acquisition should not be limited to academic purposes. However, educators should implement the e-strategies to prepare learners for professional and interpersonal communication and to enable them to apply lexical skills in authentic contexts.

The lexical competence showed potential for enhancement with the introduction of technology in vocabulary education. Booton et al. (2023) further observed that mobile-aided language learning devices with augmented reality increase learners' interest and enthusiasm, thereby enhancing vocabulary retention. Moreover, researchers might measure lexical development using quantitative tools such as lexical frequency patterns, lexical sophistication scales, and lexical diversity scales (Zare et al., 2021). These measures help educators understand how learners develop vocabulary knowledge and how it influences their communicative competence.

Lexical ability includes language skills, ways of thinking, and the use of language in communication. It includes how big, rich, and functional languages are in different situations and how they support reading, writing, speaking, and listening. However, few studies have provided empirical evidence linking lexical competence to the 4Cs in Philippine higher education, even though existing research recognizes their pedagogical and technological value. Teachers could follow this activity with vocabulary instruction to promote teamwork, innovation, and critical

thinking within the communicative approach. In this regard, lexical study will play a crucial role in educational success and work-related preparation.

Syntax

Syntax refers to the rules and patterns that determine how words are arranged to form phrases and sentences in a language. It is a core part of linguistic competence because it enables learners to construct grammatically correct and meaningful sentences. Zhou et al. (2020) explain that syntax helps speakers organize words into logical expressions, thereby making communication more effective. In learning a second language, syntactic knowledge enables students to understand complex sentences and produce accurate, fluent writing and speech.

Numerous studies have demonstrated that strong syntactic skills are essential for reading comprehension, writing proficiency, and overall language understanding. Pérez-Mayos et al. (2021) emphasized that awareness of syntactic relationships within sentences facilitates comprehension by enabling readers to process meaning more efficiently and identify textual cohesion, a claim further supported by T. Liu et al. (2023). Similarly, Y. Li and Risteski (2021) found that syntactic knowledge enhances writing accuracy and sophistication, allowing learners to organize ideas more coherently. Glavaš and Vuli´cvuli´c (2020) further noted that syntactic awareness activities—such as identifying clause boundaries, paraphrasing sentences, and analyzing authentic texts—can improve grammatical control and communicative flexibility. In addition to traditional instruction, technology-enhanced approaches have also been shown to support syntactic development. Studies by Sachan et al. (2020) and Ahmad et al. (2021) revealed that visual tools illustrating sentence structures foster metalinguistic awareness by enabling learners to independently explore syntactic patterns and understand how syntax functions in context-specific ways. Teachers can further strengthen syntactic development by providing targeted feedback on sentence variety, grammatical accuracy, and coherence (K.-H. Huang &

Chang; E. Yang et al., 2022), reinforcing syntax as a structural, functional, and creative component of language use.

Syntax is not only a linguistic structure but also a social and educational tool. Kuncoro et al. (2020) argue that syntactic proficiency enhances communicative competence by helping learners express subtle ideas in academic and professional settings. Likewise, Hutchison (2020) and Chida et al. (2020) emphasize that effective use of syntax supports higher-order skills such as argumentation, reasoning, and creativity in writing. Classroom instruction in syntax, therefore, does more than teach grammar—it also develops collaboration and critical thinking, two core elements of 21st-century learning.

Despite the growing body of international research, few Philippine studies examine how syntactic competence relates to the 4Cs—communication, collaboration, critical thinking, and creativity—particularly among college EFL learners. Future research should explore how syntactic instruction can be integrated into communicative and outcomes-based curricula to improve grammar, cognition, and overall linguistic performance.

In conclusion, grammar is a part of language and communication skills that affects how well students can read, write, and speak. There is evidence that pedagogical projects that include sentence-combining and syntactic awareness practices help students understand and write better. Teachers can use these techniques to help students learn more and become more flexible in their language use. However, few studies examine the link between teaching grammar and the skills needed for higher education in the Philippines in the 21st century. Addressing this gap can help curriculum designers and teachers determine how to teach grammar in ways that align with the needs of global conversation and language development.

Semantics

Semantics is the study of the meaning of words, phrases, and sentences. It is one of the most important parts of language because it shows how people express their thoughts and emotions (John Lyons, 2020). In higher education, semantics helps students use the right words, understand more profound meanings, and think beyond literal language in academic and professional settings (Pineda et al., 2024).

The philosophical roots of semantics still influence studies today. Kusmanto and Pulungan, 2022 explain how meaning relates to truth and interpretation, while Khitam (2023) shows that being aware of meaning helps with understanding and translation. These studies show that semantics is both a mental and structural process that connects language form, thought, and communication.

Semantics combines logic, thinking, and social experience. Classical ideas, such as those of Ibn Jinni, show that word meanings can change depending on how they are used (Muid & Mabrurrosi, 2022). Mondal (2022) explains that formal semantics is grounded in logic, while cognitive semantics is grounded in personal experience and thinking patterns. Hussien et al. (2023) say that relationships between words — such as opposites, synonyms, and categories — show how people organize vocabulary in their minds and in society. These ideas show that meaning is created through thinking and communication, not fixed beforehand. Today, studies about semantics are not only theoretical but also practical, helping improve understanding, thinking, and cultural awareness (Mantasiah Prof, 2020).

Teaching semantics helps connect meaning and structure. Juanda (2024) asserts, to construct more connections between vocabulary and communicative practices. Speaking of which, as Li et al. (2022) add that people's preferences and biases affect how they create meaning, so teachers should consider these differences. Ahyar and Yusri Ibrahim (2023) found that semantics and pragmatics work together, intertwined in translation, where meaning depends on both

linguistic accuracy and the situation. These studies support a teaching approach that helps students think critically and creatively about meaning and use it in real-life communication.

Outside the classroom, semantics also plays a role in social, emotional, and cultural communication. Papic (2023) says that meaning depends on shared understanding, not just grammar. Pritchard (2022) explains that meaning is created through interaction. Themistocleous (2024) studies how tone and rhythm express emotion in speech, while Medvid et al. (2023) show that semantic analysis can reveal power and manipulation in politics. These studies show that semantics helps people express meaning, identity, and emotion in society.

Semantics also connects with the 4Cs of learning in the 21st century: collaboration through shared meaning, critical thinking through understanding, communication through expression, and creativity through interpretation. So, semantics is not just a language skill but also a thinking and social skill that helps students understand and communicate across cultures and subjects.

Semantics continues to develop, from its philosophical roots to its modern teaching use. It includes structure, thought, and social meaning, helping people create meaning from words and experiences. However, few studies have connected semantics with 21st-century learning skills in higher education. Future research should examine how teaching semantics can enhance understanding, cultural awareness, and communication in real-life settings.

Orthography

Orthography is the study of the rules and forms of written language. It is the basis for learning to read, spell correctly, and read quickly. Readers can bridge the gap between spoken and written language by connecting pictures and sounds and understanding their meanings (Chrabaszcz et al., 2023). Most research on literacy development has focused on children. However, new research has recently turned the attention to adults, especially those who speak

English as a foreign language (EFL). Orthography is now an essential part of how well college students decode, understand, and write academic works.

Several studies have shown that knowing a lot about orthography makes reading faster and easier to understand. Johnels et al. (2024) found that there are two types of orthographic knowledge: general and word specific. They also found that both types of knowledge can predict how well a student will spell from elementary school through high school. At the college level, this result directly affects how English as a Foreign Language (EFL) is taught. Similarly, Zarić and Nagler (2021) claim that knowing how to read words and sentences correctly is a good indicator of how well you will understand what you read. They highlight the importance of diagnostic tests to find orthographic problems in adult learners. These findings show that reading and writing correctly with letters and words is important at all grade levels, especially for students learning to spell English words correctly.

Orthographic processing requires visual attention, pattern identification, and letter–sound relationship. Wong et al. (2023) found that explicit orthographic structure training increases visual discrimination and character processing in young Chinese readers, improving comprehension. K. H. Huang and Chang (2021) suggest clear orthographic teaching in early reading development, which can be used in adult and EFL literacy training. Since orthographic skills affect decoding efficiency and spelling correctness, Chen et al. (2020) provide multidimensional evaluation techniques to assess their complexity. These findings support teaching orthographic awareness, which helps students notice letter patterns, word boundaries, and sound–symbol relationships.

From the psycholinguistics perspective, there is still a strong link between understanding written language, paying attention to what you see, and moving between languages. Xu et al. (2021) found that children who have difficulty reading often have problems with orthography. This supports the idea that being aware of orthography can help kids visualize what they are reading.

Tibi et al. (2021) looked at Arabic spelling and found that the way Arabic looks has a big effect on how well you can read it. Ginestet et al. (2020) found that attention and eye movement patterns affect how well we recognize words. This suggests that attention-based techniques can help with learning letters and their sounds. Cross-linguistic studies also show that bilingual people use mental tools they already have to deal with different writing systems. Simone, Williams et al. (2021) argue that the orthographic depth of bilingualism facilitates language transmission in a productive way, thereby improving reading flexibility and comprehension.

In higher education, knowing how to write letters correctly helps with more than just reading and writing. It also helps with writing assignments and expanding your language. Lally and Rastle (2022) found that students who could see letters more clearly were able to read and copy text more accurately. McMurray (2020) supports a balanced way of teaching that includes phonological, orthographic, and morphological factors to improve reading and writing skills. In English as a foreign language (EFL) settings, written materials that offer orthographic feedback help students improve their spelling and reading fluency through computer tools that help students see words. These studies show how important it is to teach spelling in higher education so that students who speak more than one language can improve their advanced reading and writing skills.

In summary, orthographic knowledge is a difficult idea that involves understanding what you see, hear, and understand. Orthographic awareness helps with reading speed, spelling, and understanding what you read, no matter what language you speak or what amount of education you have. Research on children's literacy set the foundation for orthographic training, and recent research shows that it is still useful for adult EFL learners. To help students from a wide range of academic fields and language backgrounds, future study should look into adaptive orthographic training tools, technology-driven visual recognition systems, and cross-linguistic models.

The 4Cs Framework

Thornhill-Miller et al. (2023) assert that meaningful reform of the 4Cs Framework—Critical Thinking, Communication, Collaboration, and Creativity—is necessary to address contemporary societal shifts and changing workforce expectations. Recognized as a core model in 21st-century education, the 4Cs define key competencies that prepare learners to navigate increasingly complex and digitally mediated contexts (Partnership for 21st Century Learning (P21), 2019) The 4Cs Framework, as developed by P21 (2019) and further advanced by the Assessment and Teaching of 21st Century Skills (ATC21S) initiative, both of which advocate for adaptability, innovation, and interconnected learning environments. In support of this view, De Foo et al. (2021) emphasize that embedding the 4Cs within institutional practices reflects a commitment to cognitive adaptability, empathy, and interdisciplinary collaboration. Taken together, these perspectives position the 4Cs not merely as instructional outcomes but as essential foundations for lifelong learning and responsible global citizenship.

The most appropriate way of operationalizing the 4Cs in the classroom is by curriculum integration and inquiry-based teaching that involves the students in real-life problem-solving. Izzah and Wulandari (2023) demonstrate how the 4Cs can be designed with the help of Content and Language Integrated Learning (CLIL), where language development is incorporated into the subject of instruction with the purpose of cultivating not only communication and critical thinking skills. This is applied by Siswadi et al. (2024) to the training and education of the professions, and they find that the introduction of the 4Cs to the nursing schools evokes motivation, engagement, and active collaboration in service-learning and volunteer services. These findings confirm that the learners can be equipped with cognitive and interpersonal skills if the 4Cs are well-considered in the curricula. Further, Pardede (2020) also introduces the 4Cs into the framework of leadership

learning, where they introduce it as a model that integrates personal learning and collaborative and reflective learning outcomes, i.e., as character, competency, connection, and culture.

The 4Cs practices and measurement have been re-conceptualized by digital transformation. Technology-based instruction enables both creative thinking and collaborative work with the help of a virtual platform, online discussions, and simulation in the form of a project. Due to the effects of digital tools that facilitate creative thinking and collaborative assessment, as Thornhill-Miller et al. (2023) note, it is now possible to carry out this in a way that was not previously feasible in a conventional classroom. According to Valentijn et al. (2024), technology also increases the level of interdisciplinary interaction, which can result in learners not being limited by the constraints of disciplines or geographical areas. To the same extent, Malik et al. (2020) reveal that digital cooperation contributes to critical thinking and innovativeness of complex systems in mobility and sustainability education- a fact that can be applied to digital learning environments. So, technology is one of the means of the 4Cs; consequently, interaction, creativity, and reflective interactions among learners of various modalities are amplified.

The 4Cs assessment has been a longstanding issue since the traditional types of assessment are less sensitive to the third and fourth order thinking and collaboration. The VALUE rubrics (Valid Assessment of Learning in Undergraduate Education), Torrance Tests of Creative Thinking (TTCT), and California Critical Thinking Skills Test (CCTST) have been suggested as multidimensional tests that are consistent with performance-based results. Both De Foo et al. (2021b) and Do et al. (2023) have advocated the concept of the holistic and integrative method of evaluation that must be directed at the indicators of the individual and group level performance, and integrative systems that must offer the links between the 4Cs and the development of personal leadership and identity. Educational researchers suggest the combination of authentic

assessment in the form of digital portfolios, peer review, and collaborative projects to determine the dynamic interactions of the 4Cs in a practice-based learning process.

Altogether, the 4Cs Framework remains a cornerstone of modern educational reform. It produces flexible, creative, and entrepreneurial learners who are equipped to meet the demands of a global society. While applicable across industries, its most significant impact is in transforming classroom learning by nurturing creativity and critical thinking through inquiry, enhancing communication and technological proficiency, and fostering collaboration through problem-solving. Looking ahead, the 4Cs will continue to inform research, policy, and curriculum design, ensuring that learners are prepared to thrive in a rapidly changing and interconnected world.

Critical Thinking

Critical thinking (CT) is a high-order thinking skill, and it involves the capacity to analyze, evaluate, and synthesize information in order to make good decisions. It provides the foundation of problem-solving, creativity, and reflective judgment at school. Ali et al. (2021) and Hidayatullah et al. (2021) argue that critical thinking is not just the ability to memorize, but also the ability to reason, assess arguments, and make decisions in any sphere. Although teachers are positive about CT, Ali et al. (2021) prove that not all teachers possess comprehensive conceptual understanding and instructional methodology that could enable them to effectively apply it in classroom practice. The latter refers to the urgency of the need for professional growth, which will enable the teachers to transform their knowledge regarding CT into a systematic tool of teaching.

As recent studies demonstrate, the following teaching interventions in the development of CT are effective: problem-based learning (PBL), simulation, and metacognitive training. Hidayatullah et al. (2021) found that the students engaged in the use of PhET simulation-based learning registered improved skills in CT compared to other students in the classroom, particularly in the analysis, inference, and evaluation skills. No less significant, problem-based learning (PBL)

promotes critical thinking and reflection, which Sari and Aminatun (2021) observed in their learning environment on environmental science.

According to Chana (2022), it is also true that more competent students in CT apply deep-learning methods as compared to surface-content strategies, which means that CT, as the pure form, is dictating how students construct and internalise knowledge. In a complementary way, metacognitive interventions, such as those used in a study by, can demonstrate that self-monitoring of thinking leads to a deeper understanding of complex concepts. Sahiruddin et al. (2022) state that reading is predictive of CT proficiency, meaning intensive cognitive activities that take place due to literacy enhancement through thought processes and problem-solving.

The CT-oriented instruction results are always linked with the academic performance and learning transfer. Ali and Awan (2021) State that undergraduate science majors with the highest levels of achievements in analytical and evaluative thinking had higher academic performances, particularly in physics. This finding supports the assumption that the relationship between acquisition and application of knowledge is based on CT. Heim et al. (2022) state that undergraduate science majors with the highest levels of achievements in analytical and evaluative thinking had higher academic performances, particularly in physics. This finding supports the assumption that the relationship between acquisition and application of knowledge is based on CT.

One of the biggest challenges that has been realized in literature is the assessment of CT. Both Heim et al. (2022) and Ayu et al. (2020) mention that there are no standard structures to evaluate CT, and the majority of existing tools are not consistent and aligned with the higher levels of learning outcomes. To surmount this, researchers propose applying multidimensional rubrics and measures that are validated, such as the California Critical Thinking Skills Test (CCTST) and the VALUE rubrics that assess reasoning, interpretation, and problem-solving. Assessment models

should measure the process and outcome of thinking and not the performance level of individual students.

Despite this growing interest throughout the world, little Philippine research has been conducted on the tertiary level CT pedagogy and assessment. Other areas are not exceptions, as Jafari et al. (2020) explain, by indicating that the level of CT among university students is low in places where the curriculum is not specifically structured to teach CT. This, in the Philippine case, implies that curricular redesigning, research-based interventions, and assessment reforms that are aligned with higher-order thinking outcomes are in need. The solution to these gaps would close the gap between theory and practice in the classroom to help educators build CT as an academic skill and a life skill.

All in all, critical thinking is among the most important features of educational change and cognitive development. Such types of interventions as PBL, simulations, and metacognitive training are always evidence-based and assist in the development of analytical reasoning and learning transfer. However, the growth is hampered by definition continuity, incomplete assessment tools, and a paucity of studies on the area. It then follows that there should be one model that establishes pedagogically, outcomes, and assessment of CT, such that the learners are taught to think critically and behave rationally in a more complicated academic and professional environment.

Communication

Communication is a broad and evolving field that includes sociology, education, marketing, and technology (Littlejohn et al., 2021). Both students and professionals need good communication skills to work together, solve problems, and connect with other people in their communities. Effective communication also fosters inclusion and learning by facilitating the exchange of ideas and experiences in both local and global contexts (Afridah & Lubis, 2024). Recent

research indicates that communication is now pivotal to professional achievement, social advancement, and intercultural comprehension in the contemporary digital landscape (Aquino et al., 2024).

In both school and employment, communication involves more than just sharing information. It fosters trust, empathy, and significant connections. Masyhuri et al. (2024) stressed that communication based on local values and culture makes the relationships between school and community stronger. Lapo (2023) also said that teachers are links between schools and communities that encourage respect and collaboration. This research indicates that communication in higher education ought to transcend formal dialogue. It should help people build real relationships that make them want to learn, work together, and be active in their communities. Communication can help develop community and a shared goal when used in this manner.

Digital tools and social media have reshaped how people interact in both personal and professional settings. As Khaeriyah et al. (2023) explained that social media has transformed traditional one-way communication into a two-way, interactive exchange where people collaborate and share ideas openly. Observed that many organizations now use digital platforms to engage audiences and strengthen their reputation. However, Liu et al. (2022) warned that digital communication can lead to distractions and lower productivity when used without discipline. To address this, both students and professionals need strong digital literacy skills to use technology responsibly and effectively.

In a globalized society, intercultural communication has become increasingly important. Yang (2024) pointed out that while online slang and informal expressions make conversations more natural, they can also cause misunderstanding. Dang (2024) argued that digital platforms can act as spaces for cultural exchange, helping people build global awareness and intercultural

understanding. These findings show that digital communication can connect individuals from diverse backgrounds and bridge cultural gaps. Working in multicultural teams or diverse classrooms helps students develop intercultural competence—an essential skill for managing diversity, language barriers, and ethical communication in both digital and real-world contexts.

In today's world, communication is more than just public relations and business. It forms thoughts, builds trust, and helps make social changes. Balonas (2021) claimed that strategic communication should not only help organizations reach their goals but also help communities grow. Pieczka (2020) supports flexible ways of communicating to get people involved in politics and social issues. Setiawan et al. (2021) said that the COVID-19 outbreak made people more reliant on digital platforms, which shows the need for systems that keep people connected. Ramírez et al. (2023) added that being honest and open when talking about corporate social responsibility (CSR) is very important because it builds confidence and trust with the public. All of these studies show that communication today needs to be honest, ethical, and adaptable.

Overall, communication has become a key driver of learning, collaboration, and innovation across disciplines. Strong communication fosters innovation, creativity, and ethical responsibility. Students and professionals must be able to express ideas clearly—whether through speaking, writing, or digital media—while remaining aware of their purpose and audience. As the world becomes increasingly digital and interconnected, communication must remain inclusive, adaptable, and forward-thinking to build meaningful relationships and inspire positive social transformation (Kaur, 2020).

Collaboration

Collaboration is a concept that has gained popularity across many domains due to the growing interdependence between workplaces and schools (Francis et al., 2024). Teamwork helps individuals develop and refine problem-solving skills and contributes to productivity in both

academic and professional settings. In higher education, team-building is an effective strategy for preparing students to become cooperative workers in technologically advanced employment environments. As institutions and industries evolve, it's crucial to understand how collaboration can function effectively as a mechanism that produces innovation, inclusivity, and sustainable outcomes (X. Wang et al., 2024).

Building on this growing recognition of collaboration's importance, recent studies in tertiary education have explored how educators can intentionally design teamwork environments to cultivate creativity, problem-solving, and professional readiness among students. Teamwork in tertiary education facilitates an academic education and professional development. Klooker & Hölzle (2023) state that fostering teamwork and creativity among students requires creating a collaborative innovation environment. In their opinion, educators and institutions may develop social and physical spaces to promote active service and involvement in addressing problems within a group. On the same note, Arinaitwe (2021) discusses institutional cooperation between vocational teacher training and industry, as well as mentoring and acting on research, as means of integrating theoretical education with practical skills in the working environment. These examples show that teamwork in higher education inculcates critical thinking and flexibility, which are fundamental to school and career success.

Extending beyond the context of higher education, insights from other professional domains—particularly health care—offer valuable lessons on how professionals can enhance collaboration. The experience of other fields and health care, in particular, can contribute something valuable to learning how to establish teamwork and collaboration. Abbasi et al. (2022) focus on dental professionals' conflict management and conclude that effective communication affects group performance and outcomes. Moreover, Kida et al. (2024), found that patient safety climates are bolstered by collaborative leadership and social capital, which are also essential for

considering the contribution of teamwork to quality and safety assurance. In turn, Hossny and Sabra (2021) suggest that workplace incivility compromises cooperation between physicians and nurses, demonstrating that respect and civility are essential for successful collaboration. Jafari et al. (2020) add that interprofessional education reduces barriers to cooperation and improves the delivery of care. Based on these insights, schools can use interprofessional methods that encourage students to collaborate, take responsibility for their actions, and become more skilled professionals. The attitude of collaboration and adaptability is becoming more common as digital technology changes how people work and learn together.

In this context, technology has changed how collaboration occurs in both physical and virtual environments. Mirbabaie et al. (2021) note the potential of virtual assistants (VAs) to enhance communication and coordination within a team, especially in remote or hybrid settings where sharing information is particularly challenging. Chantziaras et al. (2021) discuss the use of collaborative robots (cobots) in industrial environments and explain how the relationship between people and machines can assist workers, making them more productive and creative as they perform more complex tasks. In the context of higher education, the identified insights point to the growing importance of digital collaboration tools, virtual exchange, and co-creation platforms that enable students to collaborate in teams without traditional boundaries.

As collaboration becomes more complex and technology-driven, it is essential to monitor its effectiveness continuously. The evaluation of the collaboration needs to be methodical, with means of capturing the process and the results. Hossain et al. (2021) review the challenges of accommodating mentally challenged employees in the workplace and emphasize that without stakeholder cooperation, success is unattainable. In the same way, Campoy-Cubillo and Jiménez Estrada (2023) develop a model for evaluating collaborative work in green manufacturing, in which dynamic models serve as the foundation for measuring the efficiency and sustainability of the

team effort. To ensure collaborative learning is inclusive, equitable, and practical, peer assessment, teamwork rubrics, and reflective evaluations can also be applied in an academic context.

Overall, the existing literature demonstrates that collaboration is essential across education, healthcare, and industry. In the event of successful collaboration, there must be favorable environments, ethical assessment practices, interprofessional learning, and digital tools. Collaborative competence, due to constant development, for students involves communication, managing diversity, and using technology through the joint creation of knowledge. The establishment of teamwork will continue to take center stage in innovation, inclusion, and long-term professional success due to the constant dynamics of workplaces and institutions.

Creativity

Creativity in education is already entering the very heart of the new research activities as teachers and scientists begin to become increasingly more interested in the importance of creativity in developing problem-solving skills, innovation, and cognitive growth (Mangubat et al., 2025). Creativity, which may be defined as the ability to generate original and useful ideas or solutions can be applied in student learning outcomes in different fields, and mathematics and science in particular (G. Baptista, 2025). The publications of modern scientists explore various forms of teaching, formative evaluation, and placement conditions that encourage creative thinking, articulating the possibilities and barriers to developing creativity in organized learning environments.(Panergayo & Pelgone, 2023).

Pedagogical strategies are quite critical in inculcating creativity, whereby the students are involved in inquiry, collaboration, and reflection. The philosophy of Problem-Based Learning (PBL) in facilitating the cultivation of creative thinking among students is also supported by Permata et al. (2022), who encourage the emergence of different solutions to environmental issues in the real world. Similarly, Fatmawati et al. (2022) cannot determine that the Creative Problem-Solving (CPS)

model encourages divergent thinking, suggesting that the instructional design of a problem directly affects creative outcomes. Khalid et al. (2020) also add that creative problem solving in mathematics is correlated with enhancing analytical skill, yet developing broader cognitive skills, not restricted to academic achievement. Jaenudin et al. (2020) verify this by showing that Project-Based Learning (PjBL) provides the balance between theory and practicality that allows students to apply creative knowledge to practical issues. All this literature proves the idea that creativity is flourishing where the teaching practices are based on inquiry, teamwork, and experiential learning.]

A crucial, albeit complex, part of the educational design is also the evaluation of creativity. Rahayuningsih et al. (2021),found that the authors of the current research emphasize the importance of the use of open-ended problems to solve, which would assess the ability of students to find and choose different paths to solutions- a strategy that is consistent with the Torrance Tests of Creative Thinking (TTCT) and other rubrics that would have assessed originality, fluency, and flexibility. Aliffia and Gusti Ketut Agung Ulupui (2023) expand on the debate by relating creativity to emotional intelligence and showing that students with balance in emotions are more flexible in thinking and possess a higher level of performance in solving problems. A combination of these studies suggests that assessment systems should include both cognitive and affective aspects of creativity to capture the full range of student innovation.

Despite the emphasis on creative learning, it is often suppressed by institutional constraints. Zhan et al. (2024) present the evidence suggesting that the inflexibility of curricula and the normative type of educational assessment system could suppress the creative powers of students, even within the system of problem-based learning. Also, Abuhadeeda et al. (2024) refer to the significance of social dynamics because the authors claim that though collaborative learning settings are linked to creativity, their performance depends on the group's co-position, diversity,

and learners' engagement. These findings imply that to encourage creativity, both effective pedagogical models and less restrictive institutional regimes that promote experimentation, inclusiveness, and flexibility are required.

Technology has provided new opportunities for fostering innovation and more access to collaborative innovation. According to Ndlela & Barnes (2024), the example of creative problem-solving can be offered, in which digital tools (e.g., creative software, online platforms, virtual collaboration environment) may be utilized to broaden the perspectives and enable new forms of interaction. This aligns with the teacher's demands to integrate digital literacy into creative pedagogy, where students should not just use technology to consume information but should design, explore, and be creative. Technology-enhanced creativity could be empowering because learning rooms are increasingly digitalized, allowing learners to become more critical in their thinking and act in finding solutions to real-world problems.

The growing number of publications signifies the leading role of creativity in modern education. Good instruction, which is problem-solving, critical, and reflective in nature, promotes creative development; creative assessment, flexible curriculum, and technology maintain it. However, these institutional barriers, which cannot be overcome immediately, remind educators that they need to re-create a learning environment where originality and exploration are appreciated. As education is dynamic, a creative strategy is essential for providing learners with greater adaptability to innovate and survive in an ever-changing world.

1.3 Theoretical Framework

This study is anchored on Dell Hymes' Theory of Communicative Competence (Hymes, 1972), which challenged the structuralist view that linguistic knowledge—particularly grammar—was the primary basis of language proficiency. Hymes (1972) argued that effective communication requires not only mastery of grammatical rules but also the ability to use language appropriately

across social contexts. He emphasized that communicative competence involves understanding when, how, and to whom language should be used, reflecting both linguistic form and sociocultural context.

Building on Hymes' sociolinguistic foundation, Canale & Swain (1980b) proposed a more systematic model consisting of four interrelated components. Grammatical competence refers to knowledge of vocabulary, syntax, morphology, and phonology, which allows learners to construct meaningful and accurate sentences. Sociolinguistic competence involves understanding social norms, cultural conventions, and context-appropriate language use. Discourse competence enables individuals to organize ideas cohesively in both spoken and written communication, while strategic competence pertains to the use of verbal and non-verbal strategies to repair breakdowns and enhance message clarity. Within this model, linguistic competence remains a core element, as it underpins learners' ability to interpret messages, produce coherent discourse, and participate successfully in communicative interactions.

To extend the discussion from linguistic ability to broader educational competencies, this study also draws on the P21 Framework for 21st Century Learning, developed by the Partnership for 21st Century Learning P21 (2019) This framework highlights the importance of the 4Cs—Critical Thinking, Communication, Collaboration, and Creativity—as essential skills required for success in contemporary academic and real-world environments. The P21 Framework positions communication not merely as the exchange of information, but as the clear articulation of ideas, active listening, and engagement in meaningful dialogue. It likewise views critical thinking as the ability to analyze information and solve problems, collaboration as coordinated interaction toward shared goals, and creativity as the generation of original and innovative ideas. These competencies collectively reflect the cognitive, social, and linguistic demands of 21st-century learning.

Integrating these perspectives, this study conceptualizes linguistic competence as the foundational resource that supports both communicative competence and the development of the 4Cs. Strong linguistic competence enhances learners' ability to think critically by enabling them to comprehend complex texts, evaluate information, and express reasoned judgments. It likewise contributes to effective collaboration through clear interaction and shared understanding, and it enriches creativity through flexible and innovative use of language. Communication—the second major construct in both Hymes' theory and the P21 Framework—serves as the bridge connecting linguistic skills to 21st-century competencies.

Thus, the Theory of Communicative Competence (Canale & Swain, 1980b; Hymes, 1972) and the Partnership for 21st Century Learning (P21) (2019) jointly provide a comprehensive foundation for the present study. Together, they explain how linguistic competence shapes students' communicative abilities and why these abilities are essential for achieving the critical thinking, collaboration, communication, and creativity skills demanded in modern education. These theoretical lenses justify examining the relationship between students' linguistic competence and their perceived 4Cs, highlighting language proficiency as both a communicative necessity and a catalyst for broader cognitive and social development.

1.4 Systematic Diagram

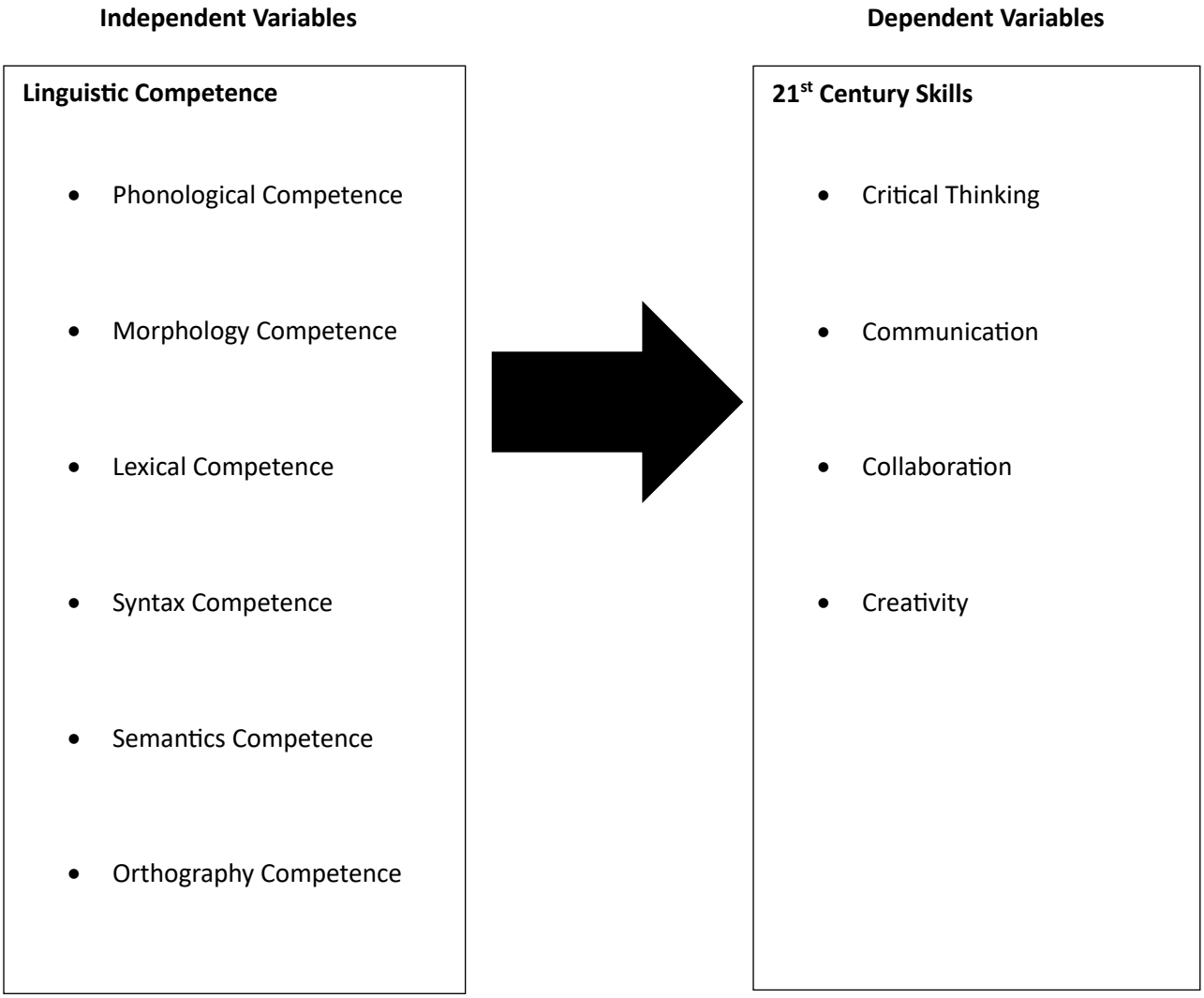


Figure 1. The schematic diagram shows the correlation between the independent and dependent variables of the sample.

1.5 Statement of the Problem/ Objectives of the Study

The study will aim to determine the relationship between 21st-century skills and linguistic competence among first-year students in the CTE and CAS departments at Saint Michael College of Caraga.

Specifically, the research seeks to address the following questions:

1. What is the level of college students' linguistic competence in terms of:

    1.1 phonology

    1.2 morphology,

    1.3 lexical,

    1.4 syntax,

    1.5 semantics; and

    1.6 orthography?

2. What is the level of 21st-century skills among college students in terms of:

    2.1 communication

    2.2 collaboration

    2.3 critical thinking; and

    2.4 creativity?

3. Is there a significant relationship between students' linguistic competence and their 21st-century skills?

1.6 Hypotheses

This study will test the following null and alternative hypotheses:

Null Hypothesis (H_o):

H_{o1}: There is no significant relationship between students' linguistic competence and communication skills.

H_{o2}: There is no significant relationship between students' linguistic competence and collaboration skills.

H_{o3}: There is no significant relationship between students' linguistic competence and creativity skills.

H_{o4}: There is no significant relationship between students' linguistic competence and critical thinking skills.

1.7 Scope and Limitation of the Study

This study will examine the relationship between linguistic competence and 21st-century competencies—specifically communication, collaboration, critical thinking, and creativity—among first-year students of the College of Teacher Education (CTE) and the College of Arts and Sciences (CAS) at Saint Michael College of Caraga during the First Semester of School Year 2025–2026.

The scope of linguistic competence in this study is limited to six components: phonology, morphology, lexical, syntax, semantics, and orthography. These areas form the basis of the researcher-made linguistic competence test administered to the participants. The study does not

include other language domains such as discourse competence, pragmatic competence, or conversational English. The study focuses solely on academic English skills.

For the 21st-century competencies, the study is confined to the four core skills identified in the P21 Framework: communication, collaboration, critical thinking, and creativity. These were measured using an adapted questionnaire developed by Varona (2020) Other 21st-century skills, such as digital literacy, media literacy, leadership, and global awareness, are not included.

The respondents of this study are limited to first-year students enrolled in selected programs under CTE and CAS. Only the following programs are included: BSED-English, BSED-Science, BSED-Math, BSED-Social Studies, BEEd, BPED, BTVTED, and BHumServ. Students from other colleges, such as Business Administration, Criminology, Computer Studies, and Maritime Education, are excluded. Only first-year students were chosen because they share common foundational language and general education courses; thus, they provide a consistent baseline for assessing linguistic competence.

The study is further limited by its use of purposive sampling, which means the results cannot be generalized to the entire SMCC population. In addition, the use of self-report questionnaires to measure the 4Cs may introduce response bias, as students may respond based on personal perceptions or social desirability. The linguistic competence test, although validated by experts, is researcher-developed and may not fully capture respondents' actual linguistic proficiency.

The study also does not account for external factors that may influence both linguistic competence and 21st-century skills, such as respondents' previous schooling, socioeconomic background, exposure to English outside the classroom, teacher effectiveness, or family environment. These variables are outside the scope of the present study.

Lastly, the research uses a descriptive–correlational design and therefore cannot establish cause-and-effect relationships. It can only identify whether a significant relationship exists between linguistic competence and the 4Cs.

1.8 Definition of Terms

To ensure conceptual precision and clarity in the use of these terms, which are operationally defined for the purposes of this study, the following terms will be used:

21st Century Competencies (4Cs). This refers to the core skills of critical thinking, communication, collaboration, and creativity, which are exemplified by the first-year respondents of the College of Teacher Education, Arts and Sciences at Saint Michael College of Caraga. These skills will be analyzed in relation to their achievement and participation in classroom tasks.

Critical Thinking. This indicates the respondents' ability to analyze, evaluate, and synthesize information in a systematic way for the purpose of making sound judgments.

Communication. This pertains to the ability of the respondents to express their ideas as well as communicate information effectively, both orally and written.

Collaboration. This refers to the respondents' ability to work together with other learners in the class. This is represented through group projects, cooperative learning activities, and other similar group work undertaken in the classroom.

Creativity. This denotes the capacity of the respondents to be creative, innovative, and imaginative in producing academic outputs and presentations.

Linguistic Competence. This captures the level of respondents' proficiency in using English in oral and written academic activities correctly and appropriately.

Phonology. This refers to the respondents' ability to recognize, distinguish, and produce the correct sounds of English, including pronunciation, stress, rhythm, and intonation.

Morphology. This refers to the respondents' ability to accurately encode and decode words using knowledge of prefixes, suffixes, and root words in a variety of academic tasks.

Lexical. This is associated with the vocabulary knowledge of the respondents and their ability to use words accurately and appropriately in their essays, discussions, and examinations.

Syntax. This refers to the respondents' command of word order and sentence structure in the construction of grammatically sound, contextually appropriate oral and written communication.

Semantics. This points to the respondents' awareness and accurate usage of word meanings, sentence and text understandings, and demonstration of such understandings in reading and in classroom conversation.

Orthography. This captures whether the respondents' written work follows the proper conventions of spelling, capitalization, and punctuation.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter presents the research methods and procedures to be employed in the study. It outlines the research design, research locale, participants, research instruments, data collection procedures, ethical considerations, and statistical techniques to be used for data analysis. The discussion explains how these methodological components will be systematically and rigorously implemented to examine the relationship between students' linguistic competence and their 21st-century competencies in accordance with the objectives of the study.

2.1 Research Design

The study will employ a descriptive–correlational quantitative research design to examine the relationship between linguistic competence and 21st-century competencies—specifically communication, collaboration, critical thinking, and creativity—among first-year students of Saint Michael College of Caraga. The descriptive component of the design will be used to determine the students' levels of linguistic competence and their self-reported 21st-century skills, as measured through a survey-based questionnaire. The correlational component will then assess whether a statistically significant relationship exists between the identified independent and dependent variables.

Descriptive statistics, including mean, standard deviation, and frequency distributions, will be utilized to summarize the students' linguistic competence scores and their perceived 4Cs. For inferential analysis, the Pearson Product–Moment Correlation Coefficient (r) will be used to determine the strength and direction of the relationship between variables when the assumptions of normality and linearity are satisfied. In cases where the data violate the normality assumption,

as determined through appropriate normality tests, Spearman's rank-order correlation (rho) will be employed as a non-parametric alternative.

This research design is appropriate because it does not involve manipulation of variables but instead examines naturally occurring relationships among measured constructs. As such, it allows the researchers to determine whether linguistic competence is significantly associated with students' self-perceived development of 21st-century skills without inferring causal relationships.

2.2 Research Materials and Methods

` This study will use two main research instruments to collect the needed data: (1) a researcher-made linguistic competence test and (2) an adapted 21st-century competencies questionnaire. These instruments will be selected and developed to ensure that they are aligned with the objectives of the study and are appropriate for first-year college students.

The linguistic competence test will be developed by the researchers to measure the respondents' proficiency in six areas of language: phonological, lexical, syntactic, semantic, morphological, and orthographic competence. The test will consist of 120 items in multiple-choice and short-response formats to assess both knowledge and application of language skills. These item types will allow the researchers to measure students' understanding and use of English in a clear and objective manner.

The 21st-century competencies questionnaire, adapted from Varona (2020), will assess respondents' perceived levels of communication, collaboration, critical thinking, and creativity. Each competency will consist of five indicators, yielding a total of 20 items that will be measured using a four-point Likert scale. The scale will be interpreted as follows: 4 – Strongly Agree**,** 3 – Agree**,** 2 – Disagree, and 1 – Strongly Disagree. Higher scores will indicate a stronger perceived manifestation of the specified 21st-century competency, while lower scores will reflect weaker perceived competence. A four-point Likert scale will be employed to eliminate a neutral response

option, thereby encouraging respondents to make more deliberate and decisive judgments regarding their perceived competencies. This approach will help reduce central tendency bias and will enhance the clarity of response patterns for statistical analysis. The adapted instrument will undergo expert validation to ensure contextual appropriateness and content relevance. Construct validity will be established through factor analysis, while internal consistency reliability will be determined using Cronbach's alpha, with coefficients ranging from 0.85 to 0.95 to be considered acceptable and indicative of high reliability.

Both instruments will be used to generate numerical data that can be analyzed using appropriate statistical tools. The use of these instruments will ensure that the data gathered are reliable, valid, and aligned with the research design of the study.

Data Gathering Procedure

The data gathering process will be conducted in an organized and systematic manner to ensure accuracy, consistency, and ethical compliance. The procedure will be divided into four major phases: (1) preparation and gathering, (2) instrument validation, (3) data collection, and (4) data consolidation and verification.

Preparation and Gathering

Before data collection, the researchers will secure written permission from the deans of the concerned colleges and approval from the appropriate institutional authorities. After obtaining approval, the researchers will coordinate with course instructors to schedule the administration of the instruments at a convenient and appropriate time.

The respondents will be informed about the purpose of the study, the procedures involved, and their rights as respondents. Informed consent forms will be distributed and collected prior to participation. Clear instructions will be given before administering the test and questionnaire to ensure that all respondents understand how to answer the instruments.

Instrument and Validation

To ensure the validity and reliability of the instruments, both tools will undergo expert validation. The linguistic competence test and the adapted 21st-century competencies questionnaire will be reviewed by three experts in English language teaching and educational research. Their feedback will focus on content accuracy, clarity of instructions, appropriateness of items, and alignment with the study's objectives.

After incorporating the validators' suggestions, a pilot test will be conducted among non-participant students to check the clarity, difficulty level, and functionality of the instruments. The reliability of the instruments will be tested using Cronbach's alpha. A reliability coefficient of 0.80 or higher will be considered acceptable, indicating that the instruments are consistent and dependable for actual data collection.

Data Collection

The administration of the linguistic competence test and the 21st-century competencies questionnaire will take place in a controlled classroom setting to maintain uniform testing conditions. The researchers will personally administer and supervise the data collection to ensure orderliness and minimize external distractions.

The respondents will be given enough time to complete both instruments. After completion, all answer sheets and questionnaires will be collected immediately to prevent loss or alteration of responses.

Data Consolidation and Verification

After data collection, all responses will be encoded into a spreadsheet or statistical software for analysis. Each response will be checked to ensure it is complete and readable. Incomplete or invalid responses will be removed following proper research guidelines.

To avoid errors, the researchers will double-check all encoded data by comparing them with the original answer sheets. Any encoding mistakes will be corrected immediately. Once verified, the final dataset will be organized and prepared for statistical analysis.

2.3 Research Locale (Research Environment)

This study will be conducted at Saint Michael College of Caraga (SMCC), formerly known as Saint Michael Institute (SMI). The institution is located at Barangay 4, Atupan Street, Nasipit, Agusan del Norte, beside Saint Michael Parish (SMP). Saint Michael College of Caraga is the only Catholic school in Nasipit and has been providing quality education for more than 75 years, making it a stable and credible academic institution for research.

Saint Michael College of Caraga will be chosen as the research locale because of its strong academic systems and quality assurance practices. The institution has passed the ISO 9001:2015 Surveillance Remote Audit, making it the first school in the Diocese of Butuan and the Caraga Region to receive ISO certification. This certification indicates that the school follows standardized academic and administrative procedures, which will help ensure an organized and reliable research environment.

The study will be conducted within two academic units of the institution: the College of Teacher Education (CTE) and the College of Arts and Sciences (CAS). These colleges will be included because their programs require students to demonstrate strong language skills and 21st-century competencies, which are the main variables of the study.

The College of Teacher Education (CTE) offers programs such as Bachelor of Secondary Education (BSED) major in English, Science, Social Studies, and Mathematics; Bachelor of Elementary Education (BEED); Bachelor of Physical Education (BPED); and Bachelor of Technical Vocational Teacher Education (BTVTED). Meanwhile, the College of Arts and Sciences (CAS) will be represented by first-year students enrolled in the Human Services program. Including these

programs will allow the study to gather relevant data from students who are exposed to language use, communication tasks, and academic interaction.

Data collection will take place in regular classroom settings within the CTE and CAS departments. These classrooms will provide a familiar and controlled academic environment where respondents can answer the research instruments independently and honestly. The researchers will ensure that the same instructions, time limits, and procedures are followed during data collection to maintain fairness and consistency.

Overall, Saint Michael College of Caraga, through the College of Teacher Education and the College of Arts and Sciences (Human Services), will provide an appropriate, credible, and academically sound research environment that supports ethical data gathering and reliable research results

2.4 Research Respondents, Population, and Sample

The population of this study will comprise all first-year students enrolled in the College of Teacher Education (CTE) and the College of Arts and Sciences (CAS) at Saint Michael College of Caraga during the First Semester of School Year 2025–2026. This population will be considered appropriate because first-year students are at a comparable academic level and are uniformly exposed to foundational courses that emphasize English language use, communication skills, and academic interaction, which are directly related to the variables examined in the study.

From this defined population, a sample of 73 first-year students will be selected using purposive sampling. This non-probability sampling technique will be employed to ensure that only students who meet specific inclusion criteria are included in the study. The inclusion criteria will require that the respondents: (1) are officially enrolled as first-year students in CTE or CAS, (2) are taking courses that involve English proficiency, communication, or language-related content, and (3) voluntarily agree to participate in the study.

The selected sample will include students from the following academic programs:

- Bachelor of Secondary Education (BSED) majors in English, Science, Mathematics, and Social Studies
- Bachelor of Elementary Education (BEEd)
- Bachelor of Physical Education (BPEd)
- Bachelor of Technical-Vocational Teacher Education (BTVTEd)
- Bachelor of Human Services (BHumServ)

Purposive sampling will be justified statistically because the primary objective of the study is not to generalize findings to the entire college population, but to examine the relationship between linguistic competence and 21st-century competencies among students who are directly exposed to language-intensive coursework. Including respondents without such exposure could weaken the accuracy of the measured relationships and introduce irrelevant variance into the data.

The sample size of 73 respondents will be considered adequate for correlational analysis, as it meets the minimum requirements for detecting moderate relationships using the Pearson Product–Moment Correlation Coefficient at a conventional level of statistical power. This sample size will allow reliable estimation of correlation coefficients while maintaining practical feasibility within the institutional context.

Prior to data collection, the researchers will obtain formal approval from the deans of the concerned colleges. All selected respondents will be informed about the purpose, procedures, and ethical safeguards of the study. Written informed consent will be secured to ensure voluntary participation, confidentiality of responses, and compliance with established ethical research standards.

2.5 Research Data and Analysis Plan

The data collected from the research instruments will be encoded, organized, and analyzed using the Statistical Package for the Social Sciences (SPSS) version 26. Descriptive statistics, such as mean and standard deviation, will be used to determine the students' levels of linguistic competence and 21st-century skills. Frequency and percentage distributions will be used to summarize the respondents' demographic information and performance levels.

To test the hypotheses, the Pearson Product-Moment Correlation Coefficient (r) will be employed to examine the strength and significance of the relationship between linguistic competence (independent variable) and the four dimensions of 21st-century skills—communication, collaboration, critical thinking, and creativity (dependent variables). The level of significance will be set at $\alpha = 0.05$. A p-value less than or equal to 0.05 will lead to the rejection of the null hypothesis; otherwise, it will be retained.

Before conducting correlation analysis, the assumptions of normality and linearity will be examined using the Shapiro–Wilk test and scatterplot inspection. If the data do not meet the assumptions for parametric testing, the Spearman's rho correlation coefficient will be used as an alternative. To complement the interpretation of the correlations, effect sizes (r) will be reported using Cohen's (1988) guidelines:

- 0.10 = small,
- 0.30 = medium,
- 0.50 = large.

Correlation strength will be described using the following conventional ranges:

- 0.00–0.19 (very weak)

- 0.20–0.39 (weak)
- 0.40–0.59 (moderate)
- 0.60–0.79 (strong)
- 0.80–1.00 (very strong)

Before analysis, all data will be checked for completeness and consistency. Missing or invalid responses will be removed using listwise deletion, provided the proportion of missing data remains minimal. All statistical results will be presented in tables and interpreted in relation to the research objectives and hypotheses.

2. 6 Ethical Standards

The researchers will strictly adhere to established ethical principles to ensure integrity, fairness, transparency, and the protection of all participants throughout the study. The following standards address all requirements related to intellectual property, informed consent, confidentiality, participant safety, data integrity, and responsible use of technology.

A. Protection of Intellectual Property Rights (IPR)

This study does not involve the creation of new technologies, products, or processes requiring patent protection. However, all academic works, research instruments, and instructional materials used are protected under copyright laws. The researchers will duly acknowledge all authors, including the 21st Century Skills Questionnaire by Varona (2020). No copyrighted or proprietary material will be reproduced, shared, or modified beyond its permitted academic use. All visual symbols, logos, and institutional identifiers used in the study will only appear with proper authorization.

B. Informed Consent

Prior to participation, all respondents will receive a clear explanation of the study's purpose, scope, procedures, expected time commitment, benefits, and minimal risks. They will be informed of their rights, including the right to withdraw at any time, skip questions they are uncomfortable with, and ask clarifying questions. Written informed consent will be secured from all respondents before data collection begins. Since the study does not involve animals or clinical human trials, additional institutional clearance for such procedures is not applicable.

C. Data Privacy and Confidentiality.

The researchers will collect only the information necessary for academic analysis and will not ask for sensitive personal details such as ethnicity, political affiliation, or medical history. All respondent identities will be anonymized using codes instead of names. Data will be stored securely in password-protected digital drives accessible only to the research team. Printed materials, if any, will be kept in locked storage. No third party will gain access to raw data unless participants give explicit written consent. Only aggregated and anonymized results will be presented in the final report.

D. Voluntary Participation and Freedom to Withdraw.

Participation in this study is entirely voluntary. Respondents may withdraw at any point without penalty, judgment, or negative academic implications. They may also skip any item they feel uncomfortable answering. No incentive or coercive element will be used to pressure participants. Only those who willingly sign the consent form will take part in the study.

E. Minimization of Harm and Risk Management.

Because the study involves only non-invasive survey questionnaires, no physical, psychological, or social harm is anticipated. Nonetheless, the researchers will remain vigilant in

ensuring respondent comfort. If any respondent experiences discomfort, distress, or a health concern during data collection, the researchers will immediately pause the procedure for that respondent. They will be allowed to rest or discontinue participation without consequence. The researchers will notify the assigned faculty member and, if necessary, assist the participant in seeking support from the school clinic or health services unit. All institutional emergency and health protocols will be strictly followed. The participant's safety, comfort, and autonomy will always take priority over the research process.

F. Beneficence and Contribution to Knowledge.

This research aims to contribute to educational improvement by examining the relationship between linguistic competence and 21st-century skills. The findings may serve as a basis for curriculum development, instructional planning, and learner support programs. Respondents involved in the study will have access to the summary results, which summarize general performance and trends of the group. This access will allow respondents to identify common strengths and weaknesses and gain insight into areas where they can focus their efforts for improvement. Individual responses will remain strictly confidential to protect privacy.

G. Justice and Fair Participant Selection.

The selection of respondents will follow transparent, fair, and non-discriminatory criteria. All eligible first-year CTE and CAS students will be invited to participate regardless of gender, socioeconomic status, religion, academic performance, or cultural background. The sampling method will ensure equal opportunity for participation and will not exclude or favor any group without valid academic justification.

H. Data Integrity and Accuracy.

The researchers will employ standardized procedures for data collection, encoding, and statistical analysis to uphold reliability and accuracy. All data will be double-checked to minimize

errors. Biases, limitations, or inconsistencies will be openly acknowledged. Under no circumstance will the researchers fabricate, manipulate, or omit data, as academic honesty and integrity remain central to the conduct of the study.

I. Transparency and Honesty in Reporting.

There will be no distortion, fabrication, or concealment of data in this study. The researchers will provide clear and detailed descriptions of the research design, methods, procedures, and limitations to ensure that the findings will be transparent, credible, and replicable. All forms of assistance received during the conduct of the research—such as expert validation of instruments, statistical computation using software, and editorial or technical support—will be properly acknowledged and not disregarded.

All materials, sources, and references used in the study will be appropriately cited in accordance with APA 7th Edition guidelines to uphold academic integrity. If artificial intelligence tools are used to assist in writing, editing, formatting, or organization of the manuscript, their role will be clearly disclosed for transparency. To address potential conflicts of interest, the researchers will disclose any affiliations, funding sources, or external influences that may affect the interpretation or reporting of results. In this study, no conflict of interest will be declared, as the research will be conducted solely for academic purposes.

J. Use of Patented or Copyrighted Materials.

Although the research does not employ patented technologies, it uses copyrighted academic instruments. For these materials, permission will be requested when required, and proper acknowledgments will be made. The researchers will ensure that the questionnaire by Varona (2020) and any other validated instruments are not distributed, modified, or reproduced outside the approved research context.

K. Ethical Considerations for Animal and Human Trials.

The study does not involve animals or laboratory-based human trials. For human participants, ethical guidelines have been strictly followed, including informed consent, voluntary participation, and minimal risk. Since the data collection involves only self-administered questionnaires, no invasive procedures or hazardous activities are included.

L. Responsible Use of AI and Related Technologies.

The researchers will utilize artificial intelligence (AI)–assisted tools solely for supportive and non-generative purposes in the conduct and documentation of the study. The AI tools that will be used include Grammarly and ChatGPT, which will contribute to improving the technical quality, clarity, and organization of the manuscript. Grammarly will be used to assist in checking grammar, spelling, sentence structure, and adherence to academic writing conventions, while ChatGPT will be used as a writing-support tool for paraphrasing, formatting, and organizing text to enhance coherence and readability. These tools will not be used to generate original research ideas, research instruments, data, statistical analyses, interpretations, conclusions, or recommendations.

Statistical analysis will be conducted using the Statistical Package for the Social Sciences (SPSS), while Microsoft Excel and Microsoft Word will be used for data processing, tabulation, and documentation. Throughout the research process, the researchers will maintain full authorship, intellectual ownership, and accountability for all aspects of the study. The use of AI and related technologies will remain transparent, ethical, and limited to technical and editorial support functions, ensuring that the integrity, credibility, and authenticity of the research are upheld.

M. Ethical Clearance and Institutional Clearance

Before the conduct of the study, the researchers will secure ethical clearance and institutional approval from the appropriate research ethics committee of Saint Michael College of Caraga. A formal request letter, approved research proposal, research instruments, and informed consent forms will be submitted for evaluation.

After approval, permission will be obtained from the deans of the College of Teacher Education (CTE) and the College of Arts and Sciences (CAS), with coordination conducted to ensure that data collection will not disrupt academic activities. Participation will be voluntary, informed consent will be obtained, and respondents' confidentiality will be strictly maintained. All data will be used solely for academic purposes, and participants will be informed of their right to withdraw at any time without penalty.

References

Abbasi, L. S., Sajjad, T., Jawed, K., & Akhtar, A. (2022). Working in Collaborative Practice: Conflict Management Styles in Dental Professionals. *Journal of the Pakistan Dental Association*, *31*(03), 131–135. https://doi.org/10.25301/JPDA.313.131

Abdon, M. M., & Barrios, A. (2022). Phonological awareness intervention in mother tongue among Filipino kindergarten learners. *Indonesian Journal of Applied Linguistics*, *12*(2), 399–411. https://doi.org/10.17509/IJAL.V12I2.45802

Abuhadeeda, M., Alaifan, A., & Hamza, E. A. (2024). *Gender and Group Preference Differentiation: Impacts on Problem Finding and Solving*. https://doi.org/10.21203/rs.3.rs-5237017/v1

Afridah, & Lubis, M. (2024). The Role of Communication and Employee Engagement in Promoting Inclusion in the Workplace: A Case Study in the Creative Industry. *Feedback International Journal of Communication*, *1*(1), 1–15. https://doi.org/10.62569/FIJC.V1I1.8

Aguion, M. A. R. (2022). *Intelligible and Comprehensible Approaches in Teaching Pronunciation in the First Years of Studying English in the Philippines*. *4*(1), 1–3. https://doi.org/10.5281/ZENODO.14811441

Ahmad, W. U., Li, H., Chang, K. W., & Mehdad, Y. (2021). Syntax-augmented Multilingual BERT for Cross-lingual Transfer. *ACL-IJCNLP 2021 - 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, Proceedings of the Conference*, *1*, 4538–4554. https://doi.org/10.18653/V1/2021.ACL-LONG.350

Ahyar, J., & Yusri Ibrahim, M. (2023). Semantic And Pragmatic Linguistic Aspects In Translation. *Jurnal Indonesia Sosial Sains*, *4*(07), 595. http://jiss.publikasiindonesia.id/

Ali, G., & Awan, R.-N. (2021). Thinking based Instructional Practices and Academic Achievement of Undergraduate Science Students: Exploring the Role of Critical Thinking Skills and

Dispositions. *Journal of Innovative Sciences*, *7*(1). https://doi.org/10.17582/journal.jis/2021/7.1.56.70

Ali, S., Qutoshi, S. B., & Jabeen, Z. (2021a). THE PERCEPTIONS AND PRACTICES OF SECONDARY SCHOOL TEACHERS FOR DEVELOPING CRITICAL THINKING SKILLS IN STUDENTS. *Pakistan Journal of Social Research*, *03*(03), 89–98. https://doi.org/10.52567/pjsr.v3i3.229

Ali, S., Qutoshi, S. B., & Jabeen, Z. (2021b). THE PERCEPTIONS AND PRACTICES OF SECONDARY SCHOOL TEACHERS FOR DEVELOPING CRITICAL THINKING SKILLS IN STUDENTS. *Pakistan Journal of Social Research*, *03*(03), 89–98. https://doi.org/10.52567/pjsr.v3i3.229

Aliffia, P., & Gusti Ketut Agung Ulupui, I. (2023). International Journal of Multidisciplinary Research and Literature PROBLEM-SOLVING APPLICATION ABILITY IN ACCOUNTING STUDENTS. *International Journal of Multidisciplinary Research and Literature IJOMRAL*, *2*(1), 1–120. https://doi.org/10.53067/ijomral.v2i1

Alkhrisheh, H. T. D., Failasofah, F., & Alkhrisheh, T. (2021). The Interrelationship among Language Use, Linguistic Competence, and Higher Order Skills. *Indonesian Research Journal in Education |IRJE|*, *5*(1), 6–22. https://doi.org/10.22437/irje.v5i1.11333

Altheneyan, A. (2025). The contribution of morphological awareness to reading comprehension among EFL college students: assessing the applicability of the morphological pathway framework. *Asian-Pacific Journal of Second and Foreign Language Education 2025 10:1*, *10*(1), 1–22. https://doi.org/10.1186/S40862-025-00354-9

Aquino, T. J. D., Cordero, A. F. F., Culla, A. K. C., Manliclic, S. N. J., Pineda, Jr., F. G., & Rafael, C. J. C. (2024). The Power of Digital Communication in Scaling Remote Engineering Teams: An Interview-Based Study on Cultural Diversity and Challenges in Telecommuting. *International Journal of Multidisciplinary: Applied Business and Education Research*, *5*(5), 1706–1715. https://doi.org/10.11594/ijmaber.05.05.19

Arinaitwe, D. (2021). Practices and strategies for enhancing learning through collaboration between vocational teacher training institutions and workplaces. *Empirical Research in Vocational Education and Training*, *13*(1). https://doi.org/10.1186/s40461-021-00117-z

Åsberg Johnels, J., Waldmann, C., & Levlin, M. (2024). Tracing the longitudinal role of orthographic knowledge in spelling development from primary to upper-secondary school. *Journal of Research in Reading*, *47*(2), 117–131. https://doi.org/10.1111/1467-9817.12443

Ayu, H. D., Saputro, S., Sarwanto, & Mulyani. (2020). *Meta-Analysis of a Blended Learning Approach: Implications for Student Critical Thinking*.

Azarcon-Sanchez, I., Chua, L., & Saranza, C. (2024). Evaluating 21st-Century Skills in English Language Teaching (ELT) in Surigao Del Norte, Philippines: A Needs Assessment for Teachers Development. *International Journal of Social Science and Human Research*, *7*(05). https://doi.org/10.47191/IJSSHR/V7-I05-50

Balonas, S. (2021). Social Change: Bringing Allies to the Field. An Interdisciplinary Model. In *Strategic Communication in Context: Theoretical Debates and Applied Research* (pp. 153–174). UMinho Editora/CECS. https://doi.org/10.21814/uminho.ed.46.8

Bollmann, M., & Søgaard, A. (2021). Error Analysis and the Role of Morphology. *EACL 2021 - 16th Conference of the European Chapter of the Association for Computational Linguistics, Proceedings of the Conference*, 1887–1900. https://doi.org/10.18653/V1/2021.EACL-MAIN.162

Booton, S. A., Hodgkiss, A., & Murphy, V. A. (2023). The impact of mobile application features on children's language and literacy learning: a systematic review. *Computer Assisted Language Learning*, *36*(3), 400–429. https://doi.org/10.1080/09588221.2021.1930057

Cabico, L. T. E., & Sales, A. J. G. (2023). Grammatical Competence Among 21st Century Tertiary Learners. *The English Teacher*, *52*(3). https://doi.org/10.52696/HECA6338

Caldoza, M. C. S., Julhamid, N. U., & Asiri, M. S. (2024). Critical Thinking Skills and Reading Proficiency among the English Language Students of Public Higher Education Institutions (HEIs) in Sulu. *Forum for Linguistic Studies*, *6*(3), 656–667. https://doi.org/10.30564/FLS.V6I3.6624

Campoy-Cubillo, M. C., & Jiménez Estrada, V. (2023). COIL PROJECTS AS A MEANS TO FOSTER COLLABORATIVE WORK IN THE PROFESSIONS. *ICERI2023 Proceedings*, *1*, 6342–6350. https://doi.org/10.21125/ICERI.2023.1579

Canale, M., & Swain, M. (1980a). Theoretical bases of communicative approaches to second language teaching and testing. *Applied Linguistics*, *1*(1), 1–47. https://doi.org/10.1093/applin/I.1.1

Canale, M., & Swain, M. (1980b). *Theoretical-Bases-of-Communicative-Approaches-to-Second-Language-Teaching-and-Testing (1)*.

Chan, A., Chen, S., Hamdani, S., Tse, B., & Cheng, K. (2023). Story telling in bilingual Urdu−Cantonese ethnic minority children: Macrostructure and its relation to microstructural linguistic skills. *Frontiers in Psychology*, *14*, 924056. https://doi.org/10.3389/FPSYG.2023.924056/BIBTEX

Chana, S. M. (2022). *International Journal of Linguistics Studies Moroccan CPGE Students' Critical Thinking Skills and Learning Approaches: The Case of Moulay Ismail CPGE Classes*. https://doi.org/10.32996/ijls

Chantziaras, G., Triantafyllidis, A., Papaprodromou, A., Chatzikonstantinou, I., Giakoumis, D., Tsakiris, A., Votis, K., & Tzovaras, D. (2021). An Augmented Reality-Based Remote Collaboration Platform for Worker Assistance. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, *12667 LNCS*, 404–416. https://doi.org/10.1007/978-3-030-68787-8_30

Chen, Y. J. I., Wilson, M., Irey, R. C., & Requa, M. K. (2020). An innovative measure of orthographic processing: Development and initial validation. *Language Testing*, *37*(3), 435–452. https://doi.org/10.1177/0265532220909310

Chida, N., Kawakoya, Y., Ikarashi, D., Takahashi, K., & Sen, K. (2020). Is stateful packrat parsing really linear in practice? A counter-example, an improved grammar, and its parsing algorithms. *CC 2020 - Proceedings of the 29th International Conference on Compiler Construction*, 155–166. https://doi.org/10.1145/3377555.3377898

Chrabaszcz, A., Gebremedhen, N. I., Alvarez, T. A., Durisko, C., & Fiez, J. A. (2023). Orthographic learning in adults through overt and covert reading. *Acta Psychologica*, *241*, 104061. https://doi.org/10.1016/J.ACTPSY.2023.104061

Dang, S. (2024). Enhancing Cross-Cultural Understanding: The Role of Digital Platforms in Cultural Exchange for Chinese Graduate Students at Northwestern University. *Journal of Social Science Humanities and Literature*, *7*(5), 55–61. https://doi.org/10.53469/jsshl.2024.07(05).10

Davydova, J. (2022). The role of social factors in the acquisition of vernacular English: A variationist study with pedagogical implications. *International Journal of Applied Linguistics*, *32*(3), 425–441. https://doi.org/10.1111/IJAL.12438

De Foo, C., Surendran, S., Jimenez, G., Ansah, J. P., Matchar, D. B., & Koh, G. C. H. (2021a). Primary care networks and starfield's 4cs: A case for enhanced chronic disease management. *International Journal of Environmental Research and Public Health*, *18*(6), 1–14. https://doi.org/10.3390/ijerph18062926

De Foo, C., Surendran, S., Jimenez, G., Ansah, J. P., Matchar, D. B., & Koh, G. C. H. (2021b). Primary Care Networks and Starfield's 4Cs: A Case for Enhanced Chronic Disease Management.

International Journal of Environmental Research and Public Health, *18*(6), 1–14. https://doi.org/10.3390/IJERPH18062926

Do, V., Maniate, J. M., Sultan, N., & Sonnenberg, L. (2023). The 4C's of influence framework: fostering leadership development through character, competence, connection and culture. *Leadership in Health Services (Bradford, England)*, *ahead-of-print*(ahead-of-print), 461–478. https://doi.org/10.1108/LHS-05-2022-0060

Dumalig, C. C., & Bayan, R. T. (n.d.). *TEACHERS' ATTITUDES TOWARDS ENGLISH PHONOLOGY IN TEACHING PRONUNCIATION AMONG ESL LEARNERS*. https://doi.org/10.5281/ZENODO.13768374

Dumanig, F. P., Gustilo, L., Thomas, E. R., An, T., & Magpale, J. (2025). Exploring Sociolectal Identity Through Speech Rhythm in Philippine English. *Languages 2025, Vol. 10, Page 101*, *10*(5), 101. https://doi.org/10.3390/LANGUAGES10050101

Duris, M., Na, I., Sonsaat-Hegelheimer, S., & Levis, J. M. (2025). Technology and Generative AI: A Revolution for Language Learning. *Pronunciation in Second Language Learning and Teaching Proceedings*, *15*(1). https://doi.org/10.31274/PSLLT.20461

Ergasheva, R. (n.d.). *THE CONCEPT OF LINGUISTIC COMPETENCE AND ITS COMPONENTS*.

Fatmawati, B., Jannah, B. M., & Sasmita, M. (2022). Students' Creative Thinking Ability Through Creative Problem Solving based Learning. *Jurnal Penelitian Pendidikan IPA*, *8*(4), 2090–2094. https://doi.org/10.29303/JPPIPA.V8I4.1846

Francis, N., Pritchard, C., Prytherch, Z., & Rutherford, S. (2024). Making teamwork work: enhancing teamwork and assessment in higher education. *FEBS Open Bio* , *15*(1), 35–47. https://doi.org/10.1002/2211-5463.13936

G. Baptista, V. E. (2025). Project-Based Learning Materials to Enhance Creative Thinking in Grade 7 Science. *International Journal of Research and Innovation in Social Science*, *IX*(IIIS), 3763–3783. https://doi.org/10.47772/IJRISS.2025.903SEDU0271

Galang, R. T., Arrozal, J. Y., Lumanlan, A. D., & Cortez, L. (2024). English Proficiency Test and Academic Performance in Purposive Communication: A Correlation Study. *International Journal of Multidisciplinary: Applied Business and Education Research*, *5*(12), 4945–4952. https://doi.org/10.11594/IJMABER.05.12.05

Giazitzidou, S., Levesque, K., & Deacon, H. (2025). Getting at How: Testing Mediating Factors in the Relation Between Morphological Awareness and Reading Comprehension in Grade 1. *Reading Research Quarterly*, *60*(2), e70007. https://doi.org/10.1002/RRQ.70007

Ginestet, E., Valdois, S., Diard, J., & Bosse, M. L. (2020). Orthographic learning of novel words in adults: effects of exposure and visual attention on eye movements. *Journal of Cognitive Psychology*, *32*(8), 785–804. https://doi.org/10.1080/20445911.2020.1823987

Glavaš, G., & Vuli´cvuli´c, I. (2020). *Is Supervised Syntactic Parsing Beneficial for Language Understanding Tasks? An Empirical Investigation*.

Heim, A. B., Walsh, C., Esparza, D., Smith, M. K., & Holmes, N. G. (2022). What influences students' abilities to critically evaluate scientific investigations? *PLoS ONE*, *17*(8 August). https://doi.org/10.1371/journal.pone.0273337

Herradura Basilan, J., & Paula Flor, B. G. (2024). *Communicative Competence Approach for Developing Alternative Learning System Learners' English Communication Skills*. https://doi.org/10.5281/ZENODO.15279129

Hidayatullah, Z., Wilujeng, I., & Munawaroh, A. (2021). Analysis of Critical Thinking Skill Through Conceptual Change Model Learning Assisted with PhET Simulation. *Proceedings of the 7th*

International Conference on Research, Implementation, and Education of Mathematics and Sciences (ICRIEMS 2020), *528*. https://doi.org/10.2991/ASSEHR.K.210305.091

Hossain, S., Moll, S., Tompa, E., & Gewurtz, R. (2021). The interactive process of negotiating workplace accommodations for employees with mental health conditions. *Work*, *69*(1), 75–90. https://doi.org/10.3233/WOR-213459

Hossny, E. K., & Sabra, H. E. (2021). Effect of nurses' perception to workplace civility climate on nurse–physician collaboration. *Nursing Open*, *8*(2), 620–627. https://doi.org/10.1002/nop2.666

Huang, K. H., & Chang, K. W. (2021). Generating Syntactically Controlled Paraphrases without Using Annotated Parallel Pairs. *EACL 2021 - 16th Conference of the European Chapter of the Association for Computational Linguistics, Proceedings of the Conference*, 1022–1033. https://doi.org/10.18653/V1/2021.EACL-MAIN.88

Huang, K.-H., & Chang, K.-W. (n.d.). *Generating Syntactically Controlled Paraphrases without Using Annotated Parallel Pairs*. https://github.com/uclanlp/synpg

Hussien, M., Faisal, A. P., & Abdallah, H. (2023). *Mohammed Hussien Mahmmoud A Cognitive Linguistic Study in Lexical Semantic Relations: Oppositeness as a Case Study*.

Hutchison, L. A. D. (2020). *Pika parsing: reformulating packrat parsing as a dynamic programming algorithm solves the left recursion and error recovery problems*. http://arxiv.org/abs/2005.06444

Hymes, D. H. (1972). *On Communicative Competence*.

Izzah, N., & Wulandari, S. (2023). Curriculum and Material Development Using Content and Language Integrated Learning (CLIL) for Teaching Science in Junior High School. *Tell : Teaching of English Language and Literature Journal*, *11*(2). https://doi.org/10.30651/tell.v11i2.18763

Jaenudin, D., Kurniasih, S., & Rizky Amalia SMA Yapida Gunung Putri Bogor Bogor, G. (2020). *Improving Students Ability in Problem Solving and Creativity Through Project-Based Learning*.

Jafari, F., Azizi, S. M., Soroush, A., & Khatony, A. (2020). Critical Thinking Level among Medical Sciences Students in Iran. In *Education Research International* (Vol. 2020). Hindawi Limited. https://doi.org/10.1155/2020/1348365

John Lyons. (2020). *Semantics*. Google Books. https://books.google.com.ph/books?id=jQA6wVLCINUC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

Juanda, J. (2024). Analysis of Language Structure and Its Implications in Modern Linguistics: A Study of the Understanding and Application of Structural Linguistic Concepts. *Journal of Educational and Social Research*, *14*(1), 226–233. https://doi.org/10.36941/jesr-2024-0019

Kadatuan, B. U., Famela Mae, U., & Limpot, M. Y. (2024). Strategy, Anxiety, and Belief in Language Learning: A Structural Equation Model in Students' Communicative Competence. In *Original Research Article Kadatuan and Limpot; Asian J. Lang. Lit. Cul. Stud* (Vol. 7, Issue 2).

Kaur, N. (2020a). Metacognitive awareness in lexical learning among malaysian students. *International Journal of English Language and Literature Studies*, *9*(3), 161–171. https://doi.org/10.18488/JOURNAL.23.2020.93.161.171

Kaur, N. (2020b). Metacognitive awareness in lexical learning among malaysian students. *International Journal of English Language and Literature Studies*, *9*(3), 161–171. https://doi.org/10.18488/JOURNAL.23.2020.93.161.171

Khaeriyah, K., Abdul Rahman, A., Amirullah, I., Gentanegara, I., Natsir, N., & Yappi Makassar, S. (2023). BRILLIANT INTERNATIONAL JOURNAL OF MANAGEMENT AND TOURISM THE EFFECT OF SOCIAL MEDIA ON COMMUNITY SOCIAL CHANGE. *BIJMT*, *3*(1). https://doi.org/10.55606/bijmt.v2i2

Khalid, M., Saad, S., Abdul Hamid, S. R., Ridhuan Abdullah, M., Ibrahim, H., & Shahrill, M. (2020). Enhancing creativity and problem solving skills through creative problem solving in teaching mathematics. *Creativity Studies*, *13*(2), 270–291. https://doi.org/10.3846/cs.2020.11027

Khan, S., & Anjum, M. A. I. (2023). Words in Mental Lexicon: A Comparative Analysis of Word Association (WA) Responses of Pakistani L1 and Afghan L2 Speakers of Urdu. *Journal of Communication and Cultural Trends*, *5*(1), 86–105. https://doi.org/10.32350/jcct.51.05

Khitam, A. K. (2023). Interrelated Complexities in The Arabic Structural Reposition. *International Journal of Arabic Language Teaching*, *5*(02), 230. https://doi.org/10.32332/ijalt.v5i02.7440

Kida, R., Fujitani, K., & Matsushita, H. (2024). Impact of Collaborative Leadership, Workplace Social Capital, and Interprofessional Collaboration Practice on Patient Safety Climate. *Journal for Healthcare Quality*, *46*(5), 268–275. https://doi.org/10.1097/JHQ.0000000000000443

Kim, Y. S. G. (2023). Executive Functions and Morphological Awareness Explain the Shared Variance between Word Reading and Listening Comprehension. *Scientific Studies of Reading*, *27*(5), 451–474. https://doi.org/10.1080/10888438.2023.2195112;PAGE:STRING:ARTICLE/CHAPTER

Klooker, M., & Hölzle, K. (2023). *A generative design of collaborative innovation space*.

Kristensen, J. K., Andersson, B., Bratlie, S. S., & Torkildsen, J. V. K. (2023). Dimensionality of Morphological Knowledge—Evidence from Norwegian Third Graders. *Reading Research Quarterly*, *58*(3), 406–424. https://doi.org/10.1002/rrq.497

Kudryashova, A. V., Rozanova, Y. V., & Sidorenko, T. V. (2020). Corpus software in EFL teaching: Examination of language exposure. *Obrazovanie i Nauka*, *22*(4), 131–145. https://doi.org/10.17853/1994-5639-2020-4-131-145

Kuncoro, A., Kong, L., Fried, D., Yogatama, D., Rimell, L., Dyer, C., & Blunsom, P. (2020). *Syntactic Structure Distillation Pretraining for Bidirectional Encoders*. https://doi.org/10.1162/tacl

Kusmanto, J., & Pulungan, A. H. (2022). THE MEANINGFULNESS OF LINGUISTIC EXPRESSIONS FROM THE TRUTH-CONDITIONAL THEORY OF MEANING (A METALINGUAL SYUDY). *Http://Jurnal.Unimed.Ac.Id/2022/Index.Php/JLT-Unimed*, 167–179. http://jurnal.unimed.ac.id/2022/index.php/JLT-Unimed

Lally, C., & Rastle, K. (2022). Orthographic and feature-level contributions to letter identification. *Quarterly Journal of Experimental Psychology (2006)*, *76*(5), 1111. https://doi.org/10.1177/17470218221106155

Lapo, P. M. (2023). About Social Communications of Nazarbayev University Library. *University Library at a New Stage of Social Communications Development. Conference Proceedings*, *2023*(8), 119–124. https://doi.org/10.15802/unilib/2023_293039

Li, C. C., Liang, H., Dong, Y., Chiclana, F., & Herrera-Viedma, E. (2022). Consistency Improvement With a Feedback Recommendation in Personalized Linguistic Group Decision Making. *IEEE Transactions on Cybernetics*, *52*(10), 10052–10063. https://doi.org/10.1109/TCYB.2021.3085760

Li, X., & Chen, T. (2024). The Role of Second Language Reading Proficiency in Moderating Second Language Word Recognition. *Education Sciences 2024, Vol. 14, Page 193*, *14*(2), 193. https://doi.org/10.3390/EDUCSCI14020193

Li, Y., & Risteski, A. (2021). The Limitations of Limited Context for Constituency Parsing. *ACL-IJCNLP 2021 - 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, Proceedings of the Conference*, *1*, 2675–2687. https://doi.org/10.18653/V1/2021.ACL-LONG.208

Liang, L., & Fryer, L. K. (2024). Phonological instruction in East Asian EFL learning: A scoping review. *System*, *123*, 103336. https://doi.org/10.1016/J.SYSTEM.2024.103336

Littlejohn, S. W. ., Foss, K. A. ., & Oetzel, J. G. . (2021). *Theories of human communication*. 503.

Liu, T., Sun, Y., Wu, J., Xu, X., Han, Y., Li, C., & Gong, B. (2023). *Unsupervised Paraphrasing under Syntax Knowledge*. www.aaai.org

Liu, X., Zheng, B., & Liu, H. (2022a). Understanding the social media interactivity paradox: the effects of social media interactivity on communication quality, work interruptions and job performance. *Information Technology and People*, *35*(7), 1805–1828. https://doi.org/10.1108/ITP-12-2020-0845

Liu, X., Zheng, B., & Liu, H. (2022b). Understanding the social media interactivity paradox: the effects of social media interactivity on communication quality, work interruptions and job performance. *Information Technology and People*, *35*(7), 1805–1828. https://doi.org/10.1108/ITP-12-2020-0845

Liu, Z., & Dou, J. (2023). Lexical density, lexical diversity, and lexical sophistication in simultaneously interpreted texts: a cognitive perspective. *Frontiers in Psychology*, *14*. https://doi.org/10.3389/fpsyg.2023.1276705

López-Pérez, M., de la Montaña Conchiña, J. L., Bravo Galán, J. L., & de la Maya Retamar, G. (2023). Bilingual Science Lexicon of Pre-Serviced Primary School Teachers. *Education Sciences 2023, Vol. 13, Page 464*, *13*(5), 464. https://doi.org/10.3390/EDUCSCI13050464

Luchini, P. L., & Kickhöfel Alves, U. (2025). The Impact of Suprasegmental Instruction on L2 English Comprehensibility and Accentedness: A Study with Argentinean Pre-Service Teachers and Brazilian Raters. *Athens Journal of Philology 2025*, *12*, 1–26. https://doi.org/10.30958/ajp.X-Y-Z

Mae, S., Lindero, E., & Sario, M. B. (2024). ISRG PUBLISHERS Teachers`Teachers`Competence and Student Learning Skills of Filipino Teachers. *Https://Isrgpublishers.Com/Isrgjahss*. https://doi.org/10.5281/zenodo.12788000

Malik, S., Swapan, M. S. H., & Khan, S. (2020). Sustainable mobility through safer roads: Translating road safety strategy into local context in western australia. *Sustainability (Switzerland)*, *12*(21), 1–20. https://doi.org/10.3390/su12218929

Mangubat, R., Calasang, V., Espina, R., Capuyan, D., Gamusa, E., Pernaa, J., Acut, D., & Garcia, M. (2025). Understanding the role of technological self-efficacy in fostering creative problem-solving and curiosity in teacher education: A structural equation modeling approach. *Journal of Technology and Science Education*, *15*(2), 456–478. https://doi.org/10.3926/JOTSE.3233

Mantasiah Prof, R. (2020). Semantic Feature Analysis Model: Linguistics Approach in Foreign Language Learning Material Development. *International Journal of Instruction*, *13*(1), 185–196. https://doi.org/10.29333/iji.2020.13112a

Mao, T. (2023). Pragmatic competence, autistic language use and the basic properties of human language. *Journal for the Theory of Social Behaviour*, *53*(3), 314–332. https://doi.org/10.1111/JTSB.12377

Martínez, R. D. I., & Bonilla, J. O. (2024). 21st century competencies: development of soft skills (communication, collaboration, critical thinking, creativity). *Revista Investigación & Praxis En CS Sociales*, *3*(1), 40–54. https://doi.org/10.24054/RIPCS.V3I1.3040

Masyhuri, M., Suud, S., & Ilyas, M. (2024). The Implementation of The Social Role of Sociology Teachers in The Community (Study of Teachers in North Lombok). *Jurnal Ilmiah Profesi Pendidikan*, *9*(1), 611–619. https://doi.org/10.29303/jipp.v9i1.2117

McMurray, S. (2020). Learning to spell for children 5–8 years of age: The importance of an integrated approach to ensure the development of phonic, orthographic and morphemic knowledge at compatible levels. *Dyslexia*, *26*(4), 442–458. https://doi.org/10.1002/dys.1663

Medvid, O., Stetsenko, O., & Azimova, O. (2023). LINGUAL MEANS AND TACTICS OF EXPRESSING SENSATIONALISM IN THE HEADLINES OF NEWS ARTICLES ON ECONOMIC SUBJECTS. *Philological Treatises*, *15*(1), 117–128. https://doi.org/10.21272/Ftrk.2023.15(1)-18

Mizan, K., Arjuna, I. H., Atiq, A. A., & Wargadinata, W. (2023). The Role of Modern Linguistics in the Learning of Arabic Language Skills. *IDEAS: Journal on English Language Teaching and Learning, Linguistics and Literature*, *11*(2), 1178–1190. https://doi.org/10.24256/ideas.v11i2.3979

Mondal, P. (2022). The Puzzling Chasm Between Cognitive Representations and Formal Structures of Linguistic Meanings. *Cognitive Science*, *46*(9), e13200. https://doi.org/10.1111/COGS.13200

Muid, A., & Mabrurrosi, M. (2022). Reflections on Ibn Jinni's Thought: Analysis of Words and Meanings in Arabic Semantics. *Al-Irfan : Journal of Arabic Literature and Islamic Studies*, *5*(1), 1–16. https://doi.org/10.36835/ALIRFAN.V5I1.5491

Muthee, M. G., Makau, M., & Amos, O. (2022). A review of techniques for morphological analysis in natural language processing. *African Journal of Science, Technology and Social Sciences*, *1*(2), 93–103. https://doi.org/10.58506/AJSTSS.V1I2.11

Mytsenko, V. I. (2024). THE ROLE OF LINGUISTIC COMPETENCE IN THE TRAINING OF STUDENTS AT HIGHER EDUCATION INSTITUTIONS. *Наукові Записки. Серія: Педагогічні Науки*, *1*(212), 116–120. https://doi.org/10.36550/2415-7988-2024-1-212-116-120

Naser, W. D., & Gandhi, N. (2022). Analysis of Students' Morphological Awareness. *Journal of Education Review Provision*, *2*(2), 76–85. https://doi.org/10.55885/JERP.V2I2.175

Ndlela, S., & Barnes, N. (2024). The Influence of Creative Problem-Solving on the Longevity of Small and Medium Enterprises. *Proceedings of The International Conference on Modern Research*

in Management, Economics and Accounting, *1*(1), 27–47. https://doi.org/10.33422/MEACONF.V1I1.174

Ngo, T. T. N., Chen, H. H. J., & Lai, K. K. W. (2024). The effectiveness of automatic speech recognition in ESL/EFL pronunciation: A meta-analysis. *ReCALL*, *36*(1), 4–21. https://doi.org/10.1017/S0958344023000113

Nishonova, K. (2024). Methodology of Developing of Pupils' Lexical Competence Through Comics in Primary English Classes. *Pubmedia Jurnal Pendidikan Bahasa Inggris*, *2*(1), 7–7. https://doi.org/10.47134/JPBI.V2I1.864

Panergayo, A. A., & Pelgone, A. J. (2023). Creative Problem-Solving in K to 12 Physics Classroom on STEM Strand. *The Normal Lights*, *17*(2), 29–56. https://doi.org/10.56278/TNL.V17I2.2174

Papic, S. (2023). CAN LINGUISTIC CORRECTNESS PROVIDE US WITH CATEGORICAL SEMANTIC NORMS? *Phenomenology and Mind*, *2023*(24), 182–191. https://doi.org/10.17454/PAM-2413

Pardede, P. (2020). Integrating the 4Cs into EFL Integrated Skills Learning. *Journal of English Teaching*, *6*(1). https://doi.org/10.33541/jet.v6i1.190

Partnership for 21st Century Learning (P21). (2019).

Pérez-Mayos, L., Carlini, R., Ballesteros, M., & Wanner, L. (2021). On the evolution of syntactic information encoded by BERT's contextualized representations. *EACL 2021 - 16th Conference of the European Chapter of the Association for Computational Linguistics, Proceedings of the Conference*, 2243–2258. https://doi.org/10.18653/V1/2021.EACL-MAIN.191

Permata, S. A. I., Sunarno, W., & Harlita, H. (2022). The Effect of the Problem Based Learning and Double Loop Problem Solving Learning Models on Problem Solving Ability in Term of Creative Thinking on Environmental Pollution Material. *Jurnal Penelitian Pendidikan IPA*, *8*(6), 2647–2653. https://doi.org/10.29303/JPPIPA.V8I6.1996

Pieczka, M. (2020). Communication and Action: Re-Reading Habermas in the Age of Activism. *Nauki o Wychowaniu. Studia Interdyscyplinarne*, *9*(2), 231–252. https://doi.org/10.18778/2450-4491.09.16

Pineda, F. G., Manliclic, S. N. J., & Rafael, C. J. C. (2024). The Power of Digital Communication in Scaling Remote Engineering Teams: An Interview-Based Study on Cultural Diversity and Challenges in Telecommuting. *International Journal of Multidisciplinary: Applied Business and Education Research*, *5*(5), 1706–1715. https://doi.org/10.11594/ijmaber.05.05.19

Pritchard, T. (2022). Proprietary linguistic meaning. *Synthese*, *200*(5), 1–20. https://doi.org/10.1007/S11229-022-03776-X/METRICS

Rahayuningsih, S., Sirajuddin, S., & Ikram, M. (2021). Using open-ended problem-solving tests to identify students' mathematical creative thinking ability. *Participatory Educational Research*, *8*(3), 285–299. https://doi.org/10.17275/PER.21.66.8.3

Ramírez, A. M., Figueroa, O. T., & Lara, J. A. S. (2023). Business Ethics and Social Responsibility. *Ciencia Latina Revista Científica Multidisciplinar*, *7*(5), 6545–6564. https://doi.org/10.37811/CL_RCM.V7I5.8242

Rodríguez, S., Neira, L. A., & Andrés, J. (2025). The impact of rhythm, intonation, and accent on the development of communicative competence in adult learners of English as a foreign language at level B1. *Pacha. Revista de Estudios Contemporáneos Del Sur Global*, *6*(18), e250417–e250417. https://doi.org/10.46652/PACHA.V6I18.417

Sachan, D. S., Zhang, Y., Qi, P., & Hamilton, W. (2020). Do Syntax Trees Help Pre-trained Transformers Extract Information? *EACL 2021 - 16th Conference of the European Chapter of the Association for Computational Linguistics, Proceedings of the Conference*, 2647–2661. https://doi.org/10.18653/v1/2021.eacl-main.228

Sahiruddin, S., Junining, E., Ubaidillah, M. F., & Len, M. C. (2022). Does Reading Correlate with Students' Critical Thinking? A Bridge to Literacy Practices. *Proceedings of the International Conference on Madrasah Reform 2021 (ICMR 2021)*, *633*, 367–373. https://doi.org/10.2991/ASSEHR.K.220104.054

Sales, A. J. (2022). Language Interference and Generative Phonology in Speech Production among Hiligaynon Native Speakers. *REiLA : Journal of Research and Innovation in Language*, *4*(3), 262–275. https://doi.org/10.31849/REILA.V4I3.7558

Sari, Y. P., & Aminatun, T. (2021). Analysis of Critical Thinking Skills on Implementation of Problem-Based Learning on Endemic Flora and Fauna in Senior High School in Bengkulu. *Proceedings of the 6th International Seminar on Science Education (ISSE 2020)*, *541*. https://doi.org/10.2991/ASSEHR.K.210326.018

Sastromiharjo, A., & Hadianto, D. (2024). Investigating eighth-grade students' lexical creativity in writing argumentative text in Indonesian language classroom. *Indonesian Journal of Applied Linguistics*, *14*(1), 173–183. https://doi.org/10.17509/IJAL.V14I1.71342

Sebahat Gözüm, Tuzcu, A., & Seher Yurt. (2020). Developing a Cultural Competency Scale for Primary Health Care Professionals. *Psikoloji Çalışmaları / Studies in Psychology*, 429–450. https://doi.org/10.26650/sp2019-0043

Setiawan, I., Disman, D., Sapriya, S., & Malihah, E. (2021). *Village Owned Enterprises (BUMDEs) Social Entrepreneurship Innovation in developing the Potential of Local Wisdom (Case Study on BUMDEs Cisantana, Kuningan Regency)*. https://doi.org/10.4108/EAI.12-12-2020.2305120

Shadiev, R., & Wang, X. (2022). A Review of Research on Technology-Supported Language Learning and 21st Century Skills. *Frontiers in Psychology*, *13*, 897689. https://doi.org/10.3389/FPSYG.2022.897689/BIBTEX

Shakirova, A. A., F.Yusupova, Z., Gordienko, O. V., & Kadyrkulova, UpelK. (2019). Development of Creativity and LinguisticMethodical Thinking of Students of Pedagogical University. *International Journal of Psychosocial Rehabilitation*, *23*(1), 163–169. https://doi.org/10.61841/JKDAN713

Shandra, N. (2021). PROCEDURE FOR COOPERATIVE PROFESSIONALLY ORIENTED WRITTEN COMMUNICATION OF PROSPECTIVE PROGRAMMERS. *Information Technologies and Learning Tools*, *84*(4), 188–210. https://doi.org/10.33407/ITLT.V84I4.4034

Shandra, N., Matviienko, L., Karpliuk, S., Povoroznyuk, R., Pochuieva, V., & Fonariuk, O. (2022). The Formation of English-Language Lexical Competence of Future Specialists of Information Technologies. *Revista Romaneasca Pentru Educatie Multidimensionala*, *14*(4), 21–39. https://doi.org/10.18662/RREM/14.4/627

Shen, L., Feng, G., Shi, L., Wu, Y., & Cao, F. (2024). The effectiveness of phonological training and morphological training in Chinese children with reading difficulty. *Reading and Writing 2024*, 1–30. https://doi.org/10.1007/S11145-024-10623-7

Shigini, P. G., & Mapunda, G. C. (2023). A morphological analysis of Kemunasukuma personal names. *Linguistik Online*, *123*(5), 95–113. https://doi.org/10.13092/LO.123.10551

Shukurullayeva, H., & Qizi, V. (2024). *The Role of Phonetics in Second Language Acquisition*. *VOLUME-5*(ISSUE-5).

Singhkum, S., & Patanasorn, C. (2022). The Association between Linguistic Competence Components and Listening Comprehension of Thai EFL Learners. *Rupkatha Journal on Interdisciplinary Studies in Humanities*, *14*(2). https://doi.org/10.21659/RUPKATHA.V14N2.09

Siswadi, Y., Saputra, B. A., & Lebdawicaksaputri, K. (2024). Correlation between the 4Cs and motivation to become volunteers among nursing students. *Healthcare in Low-Resource Settings*, *12*(3). https://doi.org/10.4081/hls.2024.12100

Sun, W. (2023). The impact of automatic speech recognition technology on second language pronunciation and speaking skills of EFL learners: a mixed methods investigation. *Frontiers in Psychology*, *14*, 1210187. https://doi.org/10.3389/FPSYG.2023.1210187/BIBTEX

Sungkur, R. K., & Shibdeen, N. (2025). Pronunciation trainer for second language learning using generative AI. *International Journal of Educational Technology in Higher Education 2025 22:1*, *22*(1), 64-. https://doi.org/10.1186/S41239-025-00561-X

Susoy, Z. (2024). Exploring the Relationship between Critical Thinking Skills and Reading Comprehension of Advanced EFL Learners. *Ahi Evran Üniversitesi Sosyal Bilimler Enstitüsü Dergisi*, *10*(3), 905–919. https://doi.org/10.31592/aeusbed.1504403

Suzukida, Y., & Saito, K. (2022). What is second language pronunciation proficiency? An empirical study. *System*, *106*. https://doi.org/10.1016/J.SYSTEM.2022.102754

Świątek, A., Pluszczyk, A., Śląski Katowicach, U., & Języka Angielskiego, I. (2023). Analiza trudności uczących się drugiego języka: przypadek polskich uczniów uczących się języka angielskiego. *Językoznawstwo*, *2/19*, 89–98. https://doi.org/10.25312/j.6864

Tafese, A. (2022). Investigation of Acquisition of Oromo Inflectional Morphology in Typically Developing Children. *International Journal of Social Science And Human Research*, *05*(05). https://doi.org/10.47191/IJSSHR/V5-I5-05

Themistocleous, C. (2024). Linguistic and Affective Prosody – A Unifying Perspective: A Systematic Review and ALE Meta-Analysis of the Cortical Organization of Prosody. *BioRxiv*, 2024.10.29.620829. https://doi.org/10.1101/2024.10.29.620829

Thornhill-Miller, B., Camarda, A., Mercier, M., Burkhardt, J. M., Morisseau, T., Bourgeois-Bougrine, S., Vinchon, F., El Hayek, S., Augereau-Landais, M., Mourey, F., Feybesse, C., Sundquist, D., & Lubart, T. (2023a). Creativity, Critical Thinking, Communication, and Collaboration: Assessment, Certification, and Promotion of 21st Century Skills for the Future of Work and Education. *Journal of Intelligence*, *11*(3). https://doi.org/10.3390/JINTELLIGENCE11030054

Thornhill-Miller, B., Camarda, A., Mercier, M., Burkhardt, J. M., Morisseau, T., Bourgeois-Bougrine, S., Vinchon, F., El Hayek, S., Augereau-Landais, M., Mourey, F., Feybesse, C., Sundquist, D., & Lubart, T. (2023b). Creativity, Critical Thinking, Communication, and Collaboration: Assessment, Certification, and Promotion of 21st Century Skills for the Future of Work and Education. *Journal of Intelligence 2023, Vol. 11, Page 54*, *11*(3), 54. https://doi.org/10.3390/JINTELLIGENCE11030054

Tibi, S., Fitton, L., & McIlraith, A. L. (2021). The development of a measure of orthographic knowledge in the Arabic language: A psychometric evaluation. *Applied Psycholinguistics*, *42*(3), 739–762. https://doi.org/10.1017/S0142716421000035

Ünaldı, İ., & Yüce, E. (2021). The Relationship Among Vocabulary Size, Grammar Proficiency, and Critical Thinking Skills of Adult Language Learners. *Adult Learning*, *32*(2), 70–78. https://doi.org/10.1177/1045159520959473

Valentijn, F. A., Schakelaar, M. Y., Hegeman, M. A., Schot, W. D., Dictus, W. J. A. G., Crnko, S., ten Broeke, T., & Bovenschen, N. (2024). A challenge-based interdisciplinary undergraduate concept fostering translational medicine. *Biochemistry and Molecular Biology Education*, *52*(2), 198–209. https://doi.org/10.1002/BMB.21804

Varona, P. (2020). 21st Century Skills: Influence on Students' Academic Performance. *SMCC Higher Education Research Journal*, *2*(1). https://doi.org/10.18868/cte.02.060120.03

Wahyuni, E., Melati, M., & Arif, N. (2022). AN ANALYSIS OF INTERCULTURAL COMMUNICATIVE COMPETENCE IN EFL TEXTBOOK OF GRADE TWELFTH OF HIGHSCHOOL. *JELT (Jambi-English Language Teaching)*, *6*(1), 31–39. https://doi.org/10.22437/JELT.V6I1.17299

Wang, X., Song, G., & Ghannam, R. (2024). Enhancing Teamwork and Collaboration: A Systematic Review of Algorithm-Supported Pedagogical Methods. In *Education Sciences* (Vol. 14, Issue 6). Multidisciplinary Digital Publishing Institute (MDPI). https://doi.org/10.3390/educsci14060675

Wang, Y., & Xie, Q. (2022). Diagnostic assessment of novice EFL learners' discourse competence in academic writing: a case study. *Language Testing in Asia*, *12*(1), 1–24. https://doi.org/10.1186/S40468-022-00197-Y/TABLES/3

Williams, G. J., Larkin, R. F., Rose, N. V., Whitaker, E., Roeser, J., & Wood, C. (2021). Orthographic Knowledge and Clue Word Facilitated Spelling in Children With Developmental Language Disorder. *Journal of Speech, Language, and Hearing Research*, *64*(10), 3909–3927. https://doi.org/10.1044/2021_JSLHR-20-00710

Wong, Y. K., Zang, X., & Inoue, T. (2023). Promoting foundational linguistic skills for reading development in young Chinese language learners: A 1-year intervention study. *Journal of Research in Reading*, *46*(3), 247–277. https://doi.org/10.1111/1467-9817.12423

Xakimova Malika Maxmud qizi. (2024). *DEVELOPMENT OF LINGUISTIC COMPETENCE IN STUDENTS*.

Xu, Q., Tao, S., Li, S., Wang, W., Li, B., & Joshi, R. M. (2021). Who are the nonresponders to intervention among Chinese children learning English as a second language? *Journal of Educational Psychology*, *113*(2), 213–229. https://doi.org/10.1037/EDU0000621

Yang, E., Bai, C., Xiong, D., Zhang, Y., Meng, Y., Xu, J., & Chen, Y. (2022). Learning Structural Information for Syntax-Controlled Paraphrase Generation. *Findings of the Association for*

Computational Linguistics: NAACL 2022 - Findings, 2079–2090. https://doi.org/10.18653/V1/2022.FINDINGS-NAACL.160

Yang, Y. (2024). Research on the Impact of Internet Language on Communication. *Communications in Humanities Research*, *46*(1), 162–166. https://doi.org/10.54254/2753-7064/46/20242298

Young-Scholten, M., Ho-Cheong Leung, A., Pavlovskaya, N., & Shepperd, L. (2024). Cross-Scriptal Orthographic Influence on Second Language Phonology. *Languages 2024, Vol. 9, Page 210*, *9*(6), 210. https://doi.org/10.3390/LANGUAGES9060210

Younis, R., & Shah, S. K. (2023). Developing Communicative Competence among the Learners: An Analysis of Secondary Level Textbooks for English in Pakistan. *Pakistan Languages and Humanities Review*, *7*(1), 456–475. https://doi.org/10.47205/PLHR.2023(7-I)42

Yu, Z., & Dong, Y. (2022). The Emergence of a Complex Language Skill: Evidence from the Self-organization of Interpreting Competence in Interpreting Students. *Bilingualism: Language and Cognition*, *25*(2), 269–282. https://doi.org/10.1017/S1366728921000870

Zare, M., Sadighi, M. S., Rassaei, F., & Bagheri, E. (2021). An Investigation of the Linguistic Complexity of IELTS Writing Topics Based on the Levels of Discourse Representation and the Degree of Meaning Coding. *Cogent Education*, *8*(1). https://doi.org/10.1080/2331186X.2020.1868235

Zarić, J., & Nagler, T. (2021). Reading comprehension on word- and sentence-level can be predicted by orthographic knowledge for German children with poor reading proficiency. *Reading and Writing*, *34*(8), 2031–2057. https://doi.org/10.1007/S11145-021-10126-9/TABLES/6

Zhan, Z., He, L., & Zhong, X. (2024). How does problem-solving pedagogy affect creativity? A meta-analysis of empirical studies. *Frontiers in Psychology*, *15*, 1287082. https://doi.org/10.3389/FPSYG.2024.1287082/BIBTEX

Zhang, H., Zhang, X., Li, M., & Zhang, Y. (2021). Morphological Awareness in L2 Chinese Reading Comprehension: Testing of Mediating Routes. *Frontiers in Psychology*, *12*, 736933. https://doi.org/10.3389/FPSYG.2021.736933/BIBTEX

Zhou, J., Li, Z., & Zhao, H. (2020). Parsing All: Syntax and Semantics, Dependencies and Spans. *Findings of the Association for Computational Linguistics Findings of ACL: EMNLP 2020*, 4438–4449. https://doi.org/10.18653/V1/2020.FINDINGS-EMNLP.398

CURRICULUM VITAE



NAME: BETHANY MAE P. MONTILLA

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Personal Information

Sex : Female

Date of birth : January 31,2005

Height : 5’0

Weight : 43

Civil Status : Single

Educational Attainment

Elementary : Nasipit Central Elementary School

Secondary : Saint Michael College of Caraga

Tertiary : N/A

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE



NAME: JANICE J. BERSAMEN

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Height : 4’11

Weight : 38

Civil Status : Single

Educational Attainment

Elementary : Nasipit Central Elementary School

Secondary : Saint Michael College of Caraga

Tertiary : N/A

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE



NAME: MARIELA L. BAGUIO

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Personal Information

Sex	:	Female
Date of birth	:	April 3, 2004
Height	:	5’4
Weight	:	43
Civil Status	:	Single

Educational Attainment

Elementary : Camagong Elementary School

Secondary : Nasipit Vocational Highschool

Tertiary : N/A

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE



NAME: CYRYL D. GALELA

ADDRESS: Purok 2, Agong-ong, Buenavista, Agusan del Norte

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Personal Information

Sex : Female

Date of birth : April 1, 2005

Height : 5’3

Weight : 66

Civil Status : Single

Educational Attainment

Elementary : Buenavista Central Elementary School

Secondary : Saint James Highschool INC

Tertiary : N/A

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE



NAME: CARLITO B. SELGAS

ADDRESS: Brgy. Talisay, Nasipit, Agusan del Norte

EMAIL: carlitoselgas70gmail.com

Personal Information

Sex : Male

Date of birth : February 1, 2005

Height : 5’2

Weight : 55

Civil Status : Single

Educational Attainment

Elementary : Talisay Central Elementary School SPED

Secondary : Saint Michael College of Caraga

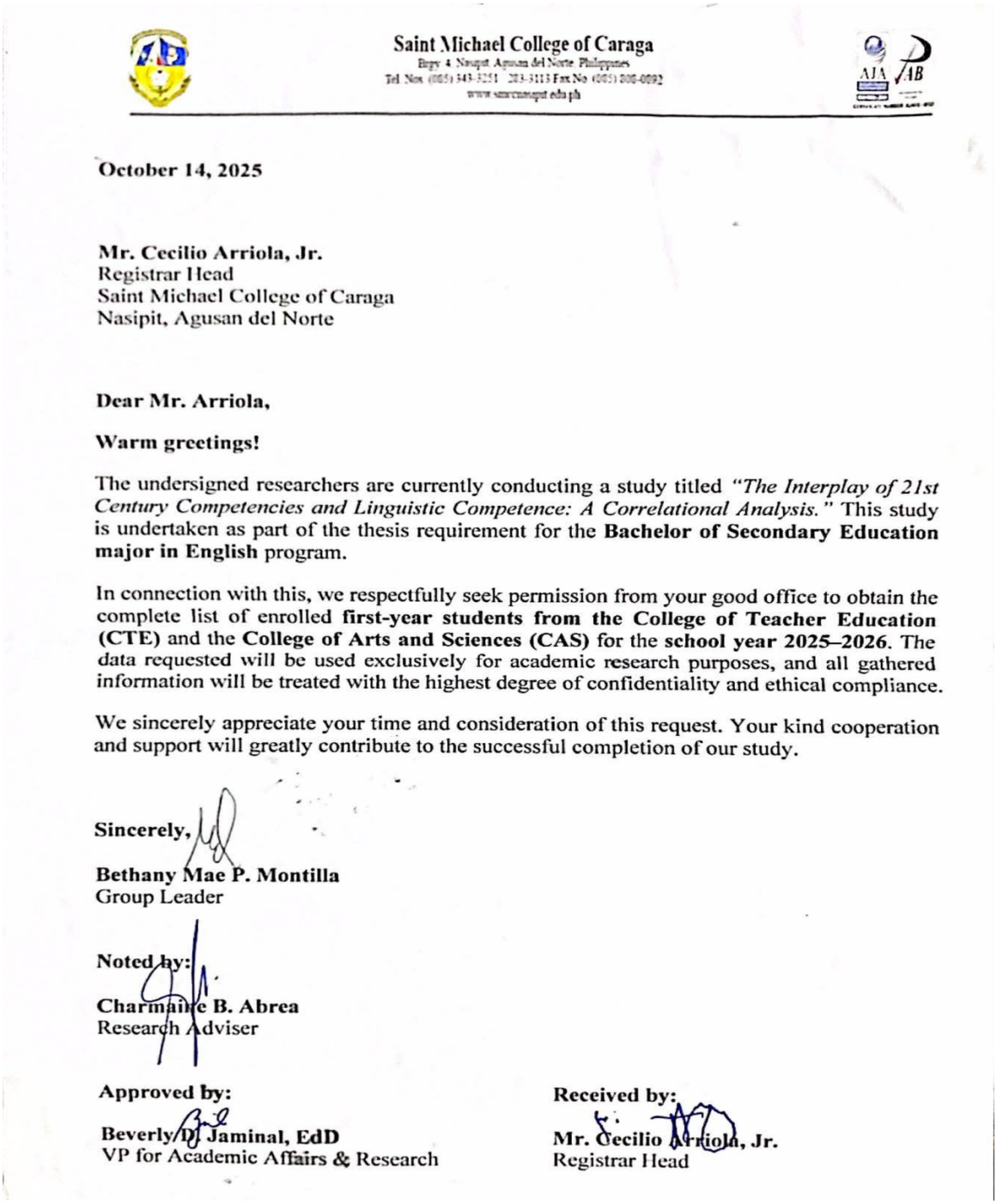
Tertiary : N/A

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

APPENDICES

A- Letter of Approval with Certificate of Appearance and Photo



Saint Michael College of Caraga
Brgy 4 Nasipit Agusan del Norte Philippines
Tel Nos (085) 343-3251 283-3113 Fax No (085) 806-0092
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October 14, 2025

Mr. Cecilio Arriola, Jr.
Registrar Head
Saint Michael College of Caraga
Nasipit, Agusan del Norte

Dear Mr. Arriola,

Warm greetings!

The undersigned researchers are currently conducting a study titled *"The Interplay of 21st Century Competencies and Linguistic Competence: A Correlational Analysis."* This study is undertaken as part of the thesis requirement for the **Bachelor of Secondary Education major in English** program.

In connection with this, we respectfully seek permission from your good office to obtain the complete list of enrolled **first-year students from the College of Teacher Education (CTE)** and the **College of Arts and Sciences (CAS)** for the **school year 2025–2026**. The data requested will be used exclusively for academic research purposes, and all gathered information will be treated with the highest degree of confidentiality and ethical compliance.

We sincerely appreciate your time and consideration of this request. Your kind cooperation and support will greatly contribute to the successful completion of our study.

Sincerely,

Bethany Mae P. Montilla
Group Leader

Noted by:

Charmaine B. Abrea
Research Adviser

Approved by:

Beverly D. Jaminal, EdD
VP for Academic Affairs & Research

Received by:

Mr. Cecilio Arriola, Jr.
Registrar Head

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October 16, 2025

Dear Ms. Varona

Warm greetings!

This letter formally requests your permission to adapt your research instrument from your study, "21st Century Skills: Influence on Students' Academic Performance". The said instrument will be utilized as part of our data-gathering tool for our research entitled "The Interplay of 21st Century Competencies and Linguistic Competence: A Correlational Analysis."

The instrument will be used to assess the relationship between linguistic competence and 21st-century skills among first-year college students from the College of Teacher Education (CTE) and the College of Arts and Sciences (CAS) at Saint Michael College of Caraga for the school year 2025–2026.

Rest assured that your original work will be properly cited and acknowledged in our study. The adapted version will be clearly identified as such and will be used solely for academic purposes.
We would greatly appreciate your kind permission to adapt and use the said instrument in our research.

Thank you very much for your time and kind consideration.

Sincerely,

Bethany Mae P. Montilla
Group Leader

Noted by:

Charmaine B. Abrea
Research Adviser

Received by:

Princess J. Varona, PhD
Head of Quality Assurance

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November 10, 2025

Melisa Maloloy-on, EdD
Dean, College of Teachers Education.
Saint Michael College of Caraga
Nasipit, Agusan del Norte

Dear Ma'am,

Warm greetings!

In partial fulfillment of the requirements for the subject Research 1, we are third-year Bachelor of Secondary Education major in English students at Saint Michael College of Caraga. On behalf of our group, I would like to ask permission from your good office to conduct a survey in the College of Teacher Education (CTE). This is for the data collection of our study entitled **"The Interplay of 21st Century Competencies and Linguistic Competence: A Correlational Analysis."** The respondents of our study are all first-year students under CTE.

Rest assured that all personal information and data will be filtered to ensure the safety and confidentiality of the respondents. Thank you and God bless!

Respectfully yours,

Bethany Mae P. Montilla
Group Leader

Noted by:

Charmaine B. Abrea
Research Adviser

Approved by:

11/10/2025
Melisa Maloloy-on, EdD
CTE Dean

EMAIL: RIID@SMCCNASIPIT.EDU.PH

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November 10, 2025

Rolly D. Llanos, MAEd
Dean, College of Arts and Sciences.
Saint Michael College of Caraga
Nasipit, Agusan del Norte

Dear Sir,

Warm greetings!

In partial fulfillment of the requirements for the subject Research 1, we are third-year Bachelor of Secondary Education major in English students at Saint Michael College of Caraga. On behalf of our group, I would like to ask permission from your good office to conduct a survey in the College of Arts and Sciences (CAS). This is for the data collection of our study entitled **"The Interplay of 21st Century Competencies and Linguistic Competence: A Correlational Analysis."** The respondents of our study are all first-year students under CAS

Rest assured that all personal information and data will be filtered to ensure the safety and confidentiality of the respondents. Thank you and God bless!

Respectfully yours,

Bethany Mae P. Montilla
Group Leader

Noted by:

Charmaine B. Abrea
Research Adviser

Approved by:

Rolly D. Llanos, MAEd
CAS Dean

EMAIL: RIID@SMCCNASIPIT.EDU.PH

Member:

B- Instrument with Informed Consent

21ST CENTURY QUESTIONNAIRE

Name (Optional): _____
Sex: ☐ Male ☐ Female
Age: _____
Course & Year: _____
Department: ☐ CTE ☐ CAS

Level of Critical Thinking Skills

	Strongly Agree	Agree	Disagree	Strongly Disagree
Questionnaire	**4**	**3**	**2**	**1**
I can understand the link connection between ideas.				
I can see the importance of ideas.				
I think of topic/issue in an objective way.				
I think independently.				
I try to look for solutions to problems.				
I evaluate generated information				
I put critical thinking skills to be used in aspects of everyday life.				
I develop my reasoning skills regularly.				
I manifest in the skillful manipulation of my own ideas/interests.				
I can compare ideas.				

Level of Communication Skills

	Strongly Agree	Agree	Disagree	Strongly Disagree
Questionnaire	**4**	**3**	**2**	**1**
I can communicate effectively with others.				
I ask questions if received messages are not clear to me.				
I convey clear meanings.				
I keep communication confidential when needed.				
I am an active listener.				
I believe being skillful in communicating provides freedom to express my thoughts/feelings properly.				

I keep communication confidential when needed.				
I use gestures to help me express my thoughts/feelings to others.				
I see to it that I interpret received messages clearly.				
I respect other people's point of view.				

Level of Collaboration Skills

	Strongly Agree	**Agree**	**Disagree**	**Strongly Disagree**
Questionnaire	**4**	**3**	**2**	**1**
I take time to respond to group problems/needs.				
I facilitate contributions of group members.				
I participate in a class project to attain a common goal.				
I contribute to group assignments.				
I foster a constructive group climate.				
I love to work when it is done by group.				
I believe that teamwork is essential at all times.				
I analyze problems without blaming others.				
I consider each one's weakness.				
I take a leadership role.				

Level of Creativity

	Strongly Agree	**Agree**	**Disagree**	**Strongly Disagree**
Questionnaire	**4**	**3**	**2**	**1**
I consider myself as a creative person.				
I am self-motivated to use my imagination.				
I like to do creative works in class.				
I explore different perspectives.				
I make connections to ideas.				
I enjoy doing creative works.				
I examine things carefully to understand what they mean.				
I organize my ideas well so that my classmates will be able to understand what I suggest.				
I have a particular time during the day when I do most of my creative works.				
I have a particular place where I do most of my creative works.				

LANGUAGE PROFICIENCY TEST

SECTION A. PHONOLOGICAL COMPETENCE (Items 1–20)

Focus: sound discrimination, word and sentence stress, rhythm, linking, intonation, and phoneme–grapheme correspondence.

(B1 Level — Items 1-5) Basic sound and stress recognition

1. (B1) Which pair of words contains the same vowel sound?

A. ship – sheep B. sit – seat C. pen – ten D. full – fool

2. (B1) Which word has a different initial sound?

A. cat B. cup C. city D. cold103.

3. (B1) Which syllable is stressed in *computer*?

A. COM B. PU C. TER D. None

4. (B1) Which word has the same final sound as *leaf*?

A. safe B. give C. laugh D. leave

5. (B1) Which pronunciation pattern is correct for the plural *books*?

A. /-s/ B. /-z/ C. /-ɪz/ D. none

(B2 Level — Items 6-13) Connected speech and stress rules

6. (B2) Which word has stress on the first syllable?

A. record (noun) B. record (verb) C. relax D. begin

7. (B2) Which word has the same vowel sound as *caught*?

A. cot B. court C. cat D. cut

8. (B2) Which sentence shows correct intonation for a yes/no question?

A. Falling tone at the end B. Rising tone at the end C. Flat tone D. Rise–fall tone

9. (B2) In natural connected speech, how is the phrase **"next year"** commonly pronounced?

A. [nek jɪər] with the /t/ sound softened or blendedB. [nekt jɪər] with a strongly pronounced /t/

C. [nexə jɪər] with an added schwa

D. [nɛk st jɪər] with a clear break between the words

10. (B2) **Which syllable is stressed in *photography*?**

A. PHO B. TO C. GRA D. PHY

11. (B2) **Which minimal-pair contrast depends on voicing?**

A. fan – van B. pin – pen C. ship – sheep D. coat – caught

12. (B2) **In the word *walked*, the final –ed is pronounced _____.**

A. /id/ B. /t/ C. /d/ D. /ed/

13. (B2) **Which word contains a silent letter?**

A. write B. rain C. rose D. rightly

(C1 Level — Items 14-20) Suprasegmental features and phonological analysis

14. (C1) Which sentence would likely have a falling intonation?

A. Where are you going? B. How beautiful this is! C. Would you like some coffee? D. Are you coming later?

15. (C1) In fluent speech, *"black coffee"* is often pronounced as:

A. [blæk kɒfi] B. [blæ kɒfi] C. [blæ kɔfi] D. [blæk ɔfi]

16. (C1) Which word pair shows difference in primary stress pattern?

A. 'import (noun)' vs im'port (verb) B. teacher vs teaching C. student vs study D. open vs opened

17. (C1) Which term refers to the reduction of unstressed vowels to /ə/?

A. assimilation B. elision C. linking D. schwa-reduction

18. (C1) In the sentence *"He didn't call, did he?"* the tag question has a ____ intonation when the speaker only wants confirmation.

A. rising B. falling C. rise-fall D. flat

19. (C1) Which process describes how sounds influence each other at boundaries (e.g., *good boy* → /gʊb bɔɪ/)?

A. elision B. assimilation C. intrusion D. reduction

20. (C1) Which pronunciation distinction differentiates *ship* /ɪ/ and *sheep* /iː/?

A. vowel length and quality B. consonant voicing C. stress placement D. syllable number

SECTION B. MORPHOLOGICAL COMPETENCE (Items 21–40)

Focus: word formation, affixes, derivation, inflection.

21. (B1) What is the base word in “unhappiness”?

A. un- B. happy C. -ness D. happiness

22. (B1) Which adjective best completes: “He is a very ____ person. He always donates time to help others”?

A. generosity B. generous C. generousness D. generate

23. (B1) Which word means “able to be believed”?

A. believable B. believe C. believing D. unbelief

24. (B1) Which past form correctly completes: “Yesterday she ____ her keys on the bus”?

A. lose B. losing C. lost D. loses

25. (B1) Which pair shows a verb-to-noun derivation?

A. decide → decision B. happy → happiness C. strong → strongly D. create → creative

26. (B2) Which word best completes: “The manager emphasized the need for greater ____ in reporting expenses”?

A. transparent B. transparency C. transparently D. transparentness

27. (B2) Which prefix forms the opposite of “legal”?

A. re- B. pre- C. il- D. sub-

28. (B2) Which adjective best completes: "The speaker delivered a highly ____ message that motivated the entire audience"?

A. inspiration B. inspire C. inspiredly D. inspiring

29. (B2) Which verb form best completes: "We must ____ the proposal before submitting it to the board"?

A. clarification B. clarifier C. clarify D. clarifiedness

30. (B2) Which noun correctly names the state/quality of "rely"?

A. reliness B. reliance C. relyment D. reliosity

31. (B2) What is the function of "-ment" in "enforcement"?

A. shows tense B. turns a verb into a noun C. shows plural D. turns a noun into a verb

32. (B2) Which negative form of "comfortable" is correct in formal writing?

A. discomfort B. uncomfort C. discomfortable D. discomfortable

33. (B2) Which form names a person who does the action: "to research → ____"?

A. researchness B. researcher C. researching D. researchedness

34. (B2) Which adjective best completes: "This policy is both reasonable and ____, which is why it gained support"?

A. implement B. implementation C. implementable D. implemented

35. (B2) Which noun is the best academic transformation of "refuse"?

A. refusion B. refusal C. refusize D. refusement

36. (C1) Which morphological breakdown of “misinterpretation” is correct?

A. mis + interpret + ation

B. mis + in + terpret + tion

C. misin + terpret + tion

D. mis + inter + pretation

37. (C1) Which sentence uses nominalization appropriately?

A. The experiment failed, which was bad.

B. The failure of the experiment raised ethical concerns.

C. The experiment failed and it was failing.

D. The experiment was a fail that failed.

38. (C1) Which adjective best completes: “The results were statistically ____”?

A. significantity B. significant C. signifying D. significance

39. (C1) Which revision shows precise morphological control of “safe/safety”?

A. make Internet use safety

B. increase the safety of Internet use

C. make Internet use safeness

D. safety the Internet

40. (C1) Which derived adjective best fits a formal tone: “Her comments were both ____ and respectful”?

A. constructive B. construction C. construct D. constructedly

SECTION C. LEXICAL COMPETENCE (Items 41-60)

Focus: vocabulary precision, collocation, register, connotation.

41. (B1) Which word best completes the sentence: “The meeting was very long and honestly quite ____; I almost fell asleep”?

A. exciting B. boring C. loud D. dangerous

42. (B1) Which word best matches the meaning “to make something better”?

A. reduce B. improve C. divide D. pretend

43. (B1) Which option shows the most natural collocation?

A. do a mistake B. make a mistake C. create a mistake D. build a mistake

44. (B1) Which word best completes the academic sentence: “This study will ____ the effects of sleep deprivation on memory”?

A. guess B. copy C. examine D. borrow

45. (B1) Which option most nearly defines “deadline”?

A. a line you cannot cross B. the final version of a story

C. a fixed time when something must be finished D. a threat from your boss

46. (B2) Which option best replaces **really good** in formal written English?

A. fairly okay B. highly effective C. kind of fine D. massively huge

47. (B2) Which word best completes the sentence: “The counselor encouraged the student to seek emotional ____ after the incident”?

A. repair B. healing C. fixing D. glue

48. (B2) Which word best completes the sentence: “The results were ____; they clearly supported our original hypothesis”?

A. confusing B. inconclusive C. predictable D. conclusive

49. (B2) Which phrasal verb best completes the sentence: “We need to ____ the main issues before we can move forward”?

A. break out B. deal with C. run over D. cut off

50. (B2) Which sentence uses an appropriate academic register?

A. The data shows that stress is kind of bad for sleep.

B. The data literally proves stress ruins sleep.

C. The data indicates a strong link between stress and poor sleep.

D. The data is crazy about sleep and stress.

51. (B2) Which sentence shows correct and natural collocation?

A. She did heavy progress this month.

B. She made heavy progress this month.

C. She made significant progress this month.

D. She did a significant progress this month.

52. (B2) Which word best completes the sentence: “The new policy is designed to ____ equal opportunities for all students”?

A. promote B. translate C. confuse D. erase

53. (B2) Which word is closest in meaning to “credible”?

A. believable B. fashionable C. forgettable D. avoidable

54. (B2) What does “tension” most nearly mean in “There was growing tension between the two departments”?

A. excitement and celebration B. physical stretching

C. stress and conflict D. silence and peace

55. (B2) Which word best completes the sentence: “The professor gave a brief ____ of the article before beginning the discussion”?

A. summary B. summarize C. summarized D. summarizing

56. (C1) Which word best fits the nuance: “Her tone during the meeting was noticeably ____”?

A. hostile B. diplomatic C. explosive D. amused

57. (C1) Which word best completes the sentence: “The study reveals a highly ____ relationship between diet and cognitive performance”?

A. random B. accidental C. insignificant D. significant

58. (C1) Which option conveys a formal, objective tone?

A. completely destroyed the other person’s argument

B. kind of won the argument

C. presented a compelling counterargument

D. roasted the other person

59. (C1) Which word best completes the sentence: “Many social media campaigns rely on emotional ____ to influence public opinion”?

A. manipulation B. homework C. translation D. recycling

60. (C1) Which word best completes the sentence: “The witness eventually ____ that parts of her original statement were inaccurate”?

A. confessed B. complained C. repeated D. pretended

SECTION D. SYNTACTIC COMPETENCE (Items 61–80)

Focus: grammar accuracy, sentence structure, agreement, subordination, cohesion.

61. (B1) Which sentence is grammatically correct?

A. He do not understand the question.

B. He does not understand the question.

C. He do not understands the question.

D. He did not understands the question.

62. (B1) Which sentence uses the correct verb form?

A. My friends was studying when I called.

B. My friend were studying when I called.

C. My friend was studying when I called.

D. My friends is studying when I called.

63. (B1) Which sentence has the correct word order for adverbs of frequency?

A. Always she is late.

B. She is always late.

C. She always is late.

D. Late always she is.

64. (B1) Which past-tense form correctly completes: “Yesterday we ____ the results of the experiment”?

A. discuss B. discussed C. discussing D. were discuss

65. (B1) Which option is a complete sentence?

A. Because the library was closed.

B. After the meeting.

C. The library was closed for repairs.

D. When I arrived at school.

66. (B2) Which sentence shows correct subject–verb agreement?

A. The number of applicants have increased.

B. The number of applicants has increased.

C. The numbers of applicant has increased.

D. The numbers of applicant have increased.

67. (B2) Which sentence maintains parallel structure?

A. She values honesty, being punctual, and responsibility.

B. She values to be honest, to be punctual, and responsibility.

C. She values honesty, punctuality, and responsibility.

D. She values honesty, to arrive on time, and responsibility.

68. (B2) Which revision best corrects the run-on?

A. The article was difficult to read; it was full of technical terms.

B. The article, was difficult to read, it was full of technical terms.

C. The article was difficult. To read, it was full. Of technical terms.

D. The article was difficult, because full of technical terms it was.

69. (B2) Which sentence uses the passive voice correctly?

A. The experiment was conduct by the assistant.

B. The experiment was conducted by the assistant.

C. The experiment conducted by the assistant.

D. The experiment has conduct by the assistant.

70. (B2) Which sentence places the modifier correctly?

A. Running to catch the bus, the folder fell out of his hands.

B. The folder fell out of his hands running to catch the bus.

C. Running to catch the bus, he dropped the folder.

D. He dropped the folder running, to catch the bus.

71. (B2) Which option maintains logical tense sequence in reported time?

A. She said she will submit the paper tomorrow.

B. She says she would submit the paper tomorrow.

C. She said she would submit the paper the next day.

D. She says she will submitted the paper the next day.

72. (B2) Which sentence is clearest and correctly punctuated with a nonessential clause?

A. The interview, which lasted two hours, it provided valuable insight.

B. The interview, which lasted two hours, provided valuable insight.

C. The interview which lasted two hours, provided valuable insight.

D. The interview which lasted two hours provided valuable insight.

73. (B2) Which verb form correctly completes the second conditional: "If the university ____ more mental health counselors, students would feel better supported"?

A. hire B. hires C. hired D. hiring

74. (B2) Which reported speech option is correct for "I am overwhelmed," she said?

A. She said that she is overwhelmed.

B. She said that she was overwhelmed.

C. She said that I was overwhelmed.

D. She said that she has overwhelmed.

75. (B2) Which sentence is NOT a run-on?

A. Many students struggle with anxiety it affects their performance.

B. Many students struggle with anxiety, it affects their performance.

C. Many students struggle with anxiety, which affects their performance.

D. Many students struggle with anxiety it, affects their performance.

76. (C1) Which sentence is the most concise and accurate for cause?

A. Due to the fact that the equipment was outdated, productivity decreased.

B. Because the equipment was outdated, productivity decreased.

C. Productivity decreased because of the fact that equipment was outdated.

D. Productivity that decreased was because the equipment was outdated.

77. (C1) Which sentence is best for academic writing and apposition?

A. The participants, most of them first-year students, were experiencing high stress.

B. The participants were first-year students, most of them, and experiencing high stress.

C. The participants, experiencing high stress, most of them first-year students.

D. Experiencing high stress, most of them, the participants were first-year students.

78. (C1) Which option uses subordination to show cause and effect?

A. The server crashed, we lost the data.

B. The server crashed because we lost the data.

C. Because the server crashed, we lost the data.

D. We lost the data, because, the server crashed.

79. (C1) Which sentence has a clear pronoun reference?

A. When the teacher met with the student, she felt calmer.

B. When the teacher met with the student, the student felt calmer.

C. When the teacher met with the student, it felt calmer.

D. When the teacher met with the student, this felt calmer.

80. (C1) Which complex sentence is grammatically correct?

A. Although many disagree, but the policy will remain in place.

B. Although many disagree, the policy will remain in place.

C. Many disagree although, the policy will remain in place.

D. Many disagree although the policy will remain in place.

SECTION E. SEMANTIC COMPETENCE (Items 81–100)

Focus: meaning, inference, figurative language, semantic relations.

81. (B1) What does “break the ice” mean in “To break the ice, the speaker told a short joke”?

A. Make the room colder B. End the event early

C. Start a conversation more comfortably D. Criticize the audience

82. (B1) Which pair consists of antonyms?

A. increase / rise B. urgent / immediate C. deny / admit D. finish / complete

83. (B1) Which sentence shows disappointment?

A. That is exactly what I hoped for!

B. This is not what I expected at all.

C. This is perfect.

D. This is better than I imagined.

84. (B1) Which sentence uses figurative (not literal) meaning?

A. The classroom was freezing.

B. The classroom was 12°C.

C. The classroom had open windows.

D. The classroom was cold because of the air conditioner.

85. (B1) Which sentence suggests possibility rather than certainty?

A. She will arrive at 8:00 p.m.

B. She is arriving at 8:00 p.m.

C. She might arrive at 8:00 p.m.

D. She arrived at 8:00 p.m.

86. (B2) What does “rebuild public trust” most likely mean after a scandal?

A. convince people to forget the past

B. regain credibility and confidence from the public

C. stop answering questions from the press

D. hide the details from reporters

87. (B2) What is implied by “I guess that is one way to solve the problem,” she said, raising an eyebrow?

A. full agreement B. solution is excellent C. skepticism or doubt D. lack of understanding

88. (B2) Which sentence signals contrast?

A. therefore, she passed B. as a result, she passed

C. however, she still failed D. consequently, she passed

89. (B2) Which option shows inference from behavior?

A. He slammed the door and did not speak to anyone for the rest of the day.

B. The door is made of wood.

C. He spoke to everyone.

D. He enjoyed the conversation.

90. (B2) Which sentence contains an idiom?

A. meet at 3 p.m. B. hit the books C. chair is wooden D. She is 20

91. (B2) Which statement best compares register?

A. The patient experienced nausea overnight.

B. The guy felt kind of sick last night.

C. Both A and B mean the same, but A is more formal.

D. Both A and B mean different things.

92. (B2) Which paraphrase best matches “Participation is mandatory”?

A. You can join if you want.

B. Joining is required.

C. You probably should join.

D. It is better not to join.

93. (B2) What is implied by “Her apology sounded rehearsed”?

A. natural and honest B. memorized and not fully sincere

C. shouted in anger D. written on paper

94. (B2) In which sentence is “literally” used correctly?

A. I literally died.

B. I was literally shaking during the earthquake.

C. That joke was literally hilarious.

D. I literally cannot even.

95. (B2) Which word best completes: “The team reached a ____ understanding, but nothing was signed”?

A. legal B. formal C. public D. mutual

96. (C1) Which tone is most accurate in “I appreciate your input, but your analysis ignores the underlying economic factors”?

A. playful B. dismissive but professional C. confused and panicked D. amused and casual

97. (C1) What does “correlation does not imply causation” mean in academic writing?

A. Co-occurrence proves cause.

B. Related variables may not cause one another.

C. Nothing is related.

D. Statistics are always wrong.

98. (C1) Which sentence contains understatement?

A. After 36 hours without sleep, he said, “I am a bit tired.”

B. I could eat ten horses.

C. The thunder was louder than a plane.

D. The fire spread quickly.

99. (C1) Which paraphrase best captures “Her leadership was instrumental in preventing the project from collapsing”?

A. She was the main reason the project survived.

B. She caused the project to fail.

C. She was not important.

D. She ignored the project.

100. (C1) What is likely implied by “The company has decided to ‘restructure,’ effective immediately”?

A. raises B. layoffs or major changes C. vacation D. nothing important

SECTION E. ORTHOGRAPHIC COMPETENCE (Items 101–120)

Focus: spelling, capitalization, punctuation, mechanics.

101. (B1) Which spelling is correct?

A. enviroment B. environment C. envirement D. enviroement

102. (B1) Which sentence uses capitalization correctly?

A. I am studying psychology at Central State University.

B. I am studying Psychology at central state university.

C. I am studying psychology at central state university.

D. I am studying Psychology at Central state University.

103. (B1) Which sentence is punctuated correctly for a simple series?

A. I bought apples, oranges and, bananas.

B. I bought apples oranges, and bananas.

C. I bought apples, oranges, and bananas.

D. I bought, apples, oranges and bananas.

104. (B1) Which option shows correct apostrophe use for singular possession?

A. The students locker was broken.

B. The student's locker was broken.

C. The students' locker was broken, it belongs to only one student.

D. The students locker' was broken.

105. (B1) Which sentence punctuates direct speech correctly?

A. “I am tired.” she said.

B. “I am tired”, she said.

C. “I am tired,” she said.

D. “I am tired.”, she said.

106. (B2) Which sentence places the comma correctly after an introductory clause?

A. After reviewing the data the committee reached a decision.

B. After reviewing the data, the committee reached a decision.

C. After reviewing the data the committee, reached a decision.

D. After reviewing, the data the committee reached a decision.

107. (B2) Which spelling is correct?

A. occassion B. ocassion C. occasion D. occation

108. (B2) Which sentence capitalizes titles/institutions correctly?

A. The Dean of Students will meet with the president of the psychology club.

B. The dean of students will meet with the President of the Psychology Club.

C. The Dean of students will meet with the President of the Psychology club.

D. The dean of Students will meet with the president of the Psychology Club.

109. (B2) Which sentence uses a semicolon correctly?

A. The study was complex; however, the findings were clear.

B. The study was complex; because the findings were clear.

C. The study was complex; and the findings were clear.

D. The study was complex; the findings were, clear.

110. (B2) Which sentence hyphenates a compound adjective correctly before a noun?

A. a well researched-report

B. a well-researched report

C. a well researched report

D. a well-research report

111. (B2) Which sentence punctuates a series of three items correctly?

A. planning, funding and, community support

B. planning, funding, and community support

C. requires, planning, funding, and community support

D. planning funding, and community support

112. (B2) Which option best corrects the comma splice: “The survey was anonymous, the responses were honest”?

A. The survey was anonymous; the responses were honest.

B. The survey was anonymous, but, the responses were honest.

C. The survey was anonymous the responses were honest.

D. The survey was anonymous: and the responses were honest.

113. (B2) Which sentence uses quotation punctuation correctly (AmE)?

A. …is “a double-edged sword”.

B. …is “a double-edged sword.”

C. …is, “a double-edged sword”.

D. …, that social media is “a double-edged sword”.

114. (B2) Which sentence shows consistent tense and correct mechanics?

A. She emails the professor yesterday…

B. She emailed the professor yesterday…

C. She email the professor yesterday…

D. She emailing the professor yesterday…

115. (B2) Which sentence uses the apostrophe correctly for plural possession?

A. The students’ concerns were addressed by the dean.

B. The student's concerns were addressed … referring to a group.

C. The students concerns' were addressed…

D. The students concern's were addressed…

116. (C1) Which sentence uses an em dash correctly for emphasis?

A. The results were clear — the treatment group improved significantly.

B. The results were — clear the treatment group improved significantly.

C. The results were clear the treatment group — improved significantly.

D. The results — were clear the treatment group improved significantly.

117. (C1) Which sentence uses a colon correctly to introduce a list?

A. The participants reported three main sources of stress: exams, finances, and sleep deprivation.

B. The participants reported: three main sources of stress…

C. The participants: reported three main sources of stress…

D. …three main sources of stress, exams: finances, and sleep deprivation.

118. (C1) Which sentence punctuates two independent clauses most appropriately for academic writing?

A. Anxiety affects sleep it also affects concentration.

B. Anxiety affects sleep; it also affects concentration.

C. Anxiety affects sleep, it also affects concentration.

D. Anxiety affects sleep also affects concentration.

119. (C1) Which sentence correctly sets off nonessential information with commas?

A. The professor, who has published five books, will lead the seminar.

B. The professor who has published five books will lead the seminar.

C. The professor who, has published five books will lead the seminar.

D. The professor who has published, five books will lead the seminar.

120. (C1) Which sentence shows final-edit quality in mechanics, clarity, and tone?

A. proves … bad for everyone…

B. kind of shows … pretty harmful

C. suggests that excessive social media use may negatively affect concentration

D. totally destroying attention spans

MASTER ANSWER KEY

Phonological Competence (1-20): 1 C, 2 C, 3 B, 4 A, 5 A, 6 A, 7 B, 8 B, 9 A, 10 C, 11 A, 12 B, 13 A, 14 B, 15 A, 16 A, 17 D, 18 B, 19 B, 20 A

Morphological (21–40): 21 B, 22 B, 23 A, 24 C, 25 A, 26 B, 27 C, 28 D, 29 C, 30 B, 31 B, 32 A, 33 B, 34 C, 35 B, 36 A, 37 B, 38 B, 39 B, 40 A

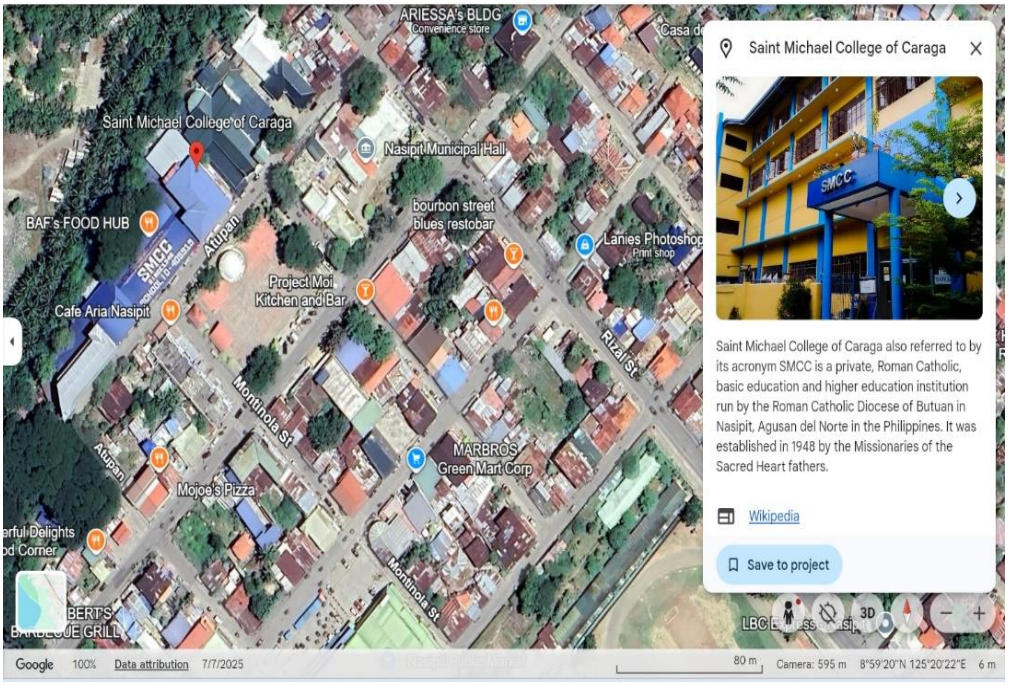
Lexical (41–60): 41 B, 42 B, 43 B, 44 C, 45 C, 46 B, 47 B, 48 D, 49 B, 50 C, 51 C, 52 A, 53 A, 54 C, 55 A, 56 B, 57 D, 58 C, 59 A, 60 A

Syntactic (61–80): 61 B, 62 C, 63 B, 64 B, 65 C, 66 B, 67 C, 68 A, 69 B, 70 C, 71 C, 72 B, 73 C, 74 B, 75 C, 76 B, 77 A, 78 C, 79 B, 80 B

Semantic (81–100): 81 C, 82 C, 83 B, 84 A, 85 C, 86 B, 87 C, 88 C, 89 A, 90 B, 91 C, 92 B, 93 B, 94 B, 95 D, 96 B, 97 B, 98 A, 99 A, 100 B

Orthographic (101–120): 101 B, 102 A, 103 C, 104 B, 105 C, 106 B, 107 C, 108 A, 109 A, 100 B, 111 B, 112 A, 113 B, 114 B, 115 A, 116 A, 117 A, 118 B, 119 A, 120 C

C- Map of the Research Locale (Include Coordinates)



D- Narrative & Photo Documentation

I. Formulation and Checking of Titles



This photo shows our research group working together to formulate and check possible titles for our study. Using our laptops and notes, we discussed and refined ideas to ensure clarity, relevance, and alignment with the purpose of the research. This activity highlights our collaboration and teamwork during the initial stage of the research process.

II. Title Hearing



This photo captures our research group during the title hearing, where we formally presented and discussed our proposed research title. Panel members provided feedback and suggestions to improve clarity, focus, and relevance, helping us refine the title before proceeding to the next phase of the study.

III. Formulation of Chapter 1-2



This photo captures our research group during the formulation of Chapters 1 and 2 of the study. We collaboratively discussed the background, statement of the problem, and related literature to ensure alignment with the research objectives. This process helped strengthen the foundation and direction of the study before proceeding to data collection.

IV. Submitting Letter for Request for Respondent



This photo shows our research group during the submission of a formal request letter to seek permission for conducting the study. The purpose and scope of the research were clearly explained to the concerned authority. This step ensured compliance with ethical standards and institutional requirements prior to gathering data.

V. Checking Chapter 1-2



This photo documents the researchers' consultation with their adviser during the review of Chapters 1 and 2, where feedback and guidance were provided to improve clarity, coherence, and alignment with the approved research title.

VI. Chapter 1-2 Proposal

E- Ethics Committee Approval

CERTIFICATE OF AUTHENTIC AUTHORSHIP

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